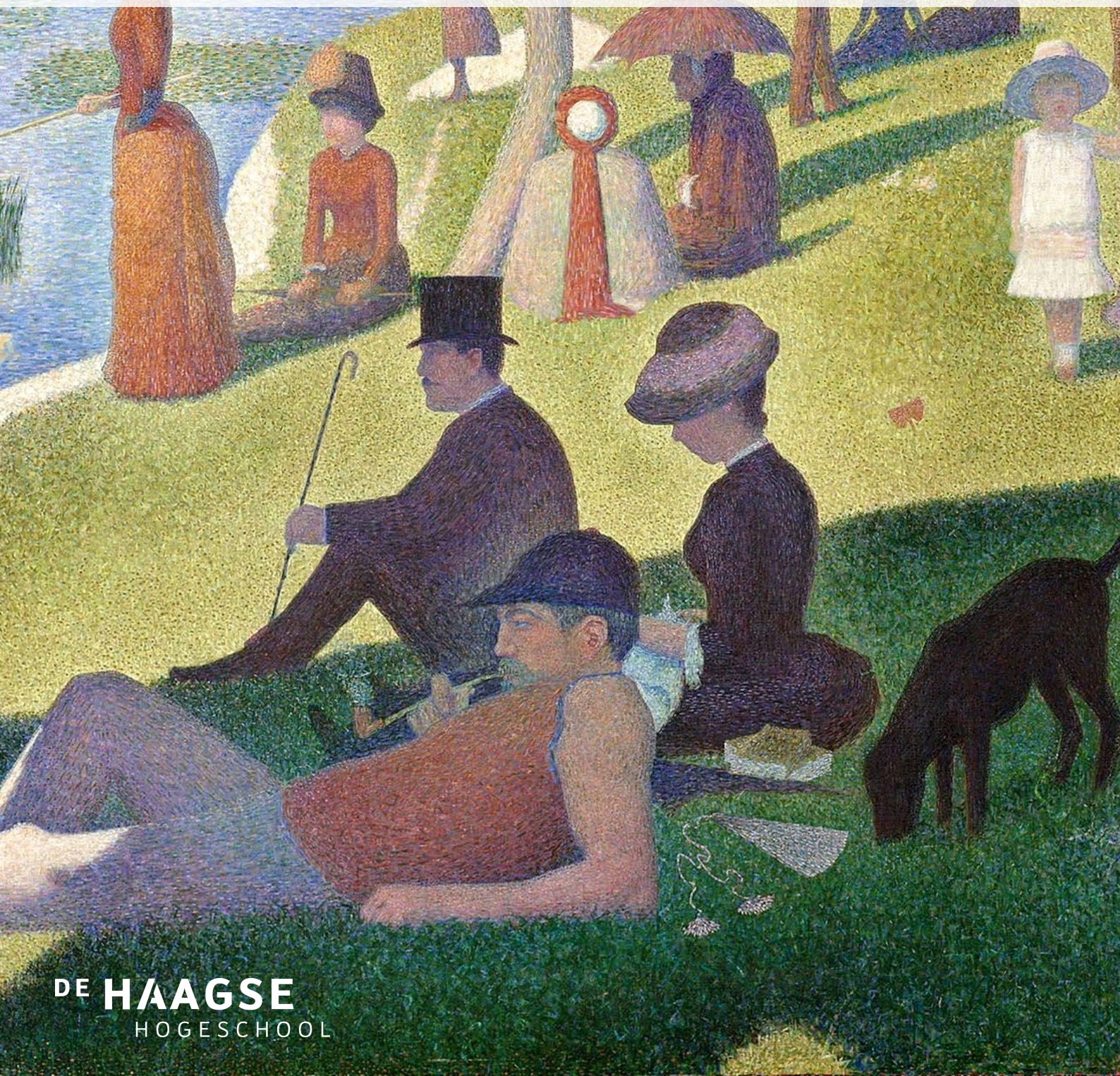


NO FAIRNESS WITHOUT AWARENESS

Applied Research into Equity in Higher Education
Inaugural Address upon Appointment
as Professor of Learning Technology & Analytics

T.C. BAKKER, PHD



NO FAIRNESS WITHOUT AWARENESS

Applied Research into Equity in Higher Education
Inaugural Address upon Appointment
as Professor of Learning Technology & Analytics

T.C. BAKKER, PHD

The research presented in this inaugural address was conducted by the research group Learning Technology & Analytics of the Centre of Expertise Global & Inclusive Learning and the Faculty of IT & Design of The Hague University of Applied Sciences.

DOI: <https://doi.org/10.5281/zenodo.18669524>

Parts of this inaugural address were delivered on November 21, 2024, at The Hague University of Applied Sciences. The entire meeting can be viewed online on the THUAS website.

Cover: *A Sunday afternoon on the island of La Grande Jatte*, George Seurat, source: Wikipedia

© T.C. Bakker, PhD, The Hague, 2024

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without the prior written permission of the author.

THE HAGUE UNIVERSITY OF APPLIED SCIENCES

NO FAIRNESS WITHOUT AWARENESS

Applied Research into Equity in Higher Education

Inaugural Address upon Appointment

as Professor of Learning Technology & Analytics

by

T.C. Bakker, PhD

*Footfalls echo in the memory
Down the passage we did not take
Towards the door we never opened.*

— T.S. Elliot

The accommodation of differences is the essence of true equality

— Will Kymlicka

Dedicated to
Coby Breur, 1941–2013,
and Gerrit Bakker, 1936–2017

CONTENT

I INTRODUCTION – RESEARCH INTO EQUITY AND FAIRNESS – CONCLUSIONS

1	INTRODUCTION	1
1.1	Learning analytics, Educational Data Mining, and Social Justice	2
1.2	Purpose and structure of inaugural address	4
2	EQUITY IN HIGHER EDUCATION	7
2.1	Personal motivation for equity	7
2.2	The importance of higher education for equity	8
2.3	Limitations in thinking about equity and higher education	8
3	ANALYSIS OF BIAS AND FAIRNESS IN STUDY DATA	11
3.1	Diversity in intake and outflow	11
3.2	Concerns about equity and bias in study data	12
3.3	Sensitive characteristics versus special personal data	14
3.4	Current research into fairness in (higher) education	15
3.5	Fairness analysis	16
4	CONCLUSIONS, RECOMMENDATIONS, AND FOLLOW-UP RESEARCH	25
4.1	The main findings of the fairness study	25
4.2	Applications of this research	27
4.3	Follow-up research	31
	SUMMARY	33
	ACKNOWLEDGEMENTS	37
	REFERENCES	39

II APPENDICES

A	APPENDIX A - FAIRNESS ANALYSIS PROCESS	45
A.1	Fairness analysis	45
B	APPENDIX B - FAIRNESS RATIOS	55
B.1	Accuracy Equality	55

B.2	Equal Opportunity	56
B.3	Predictive Equality	57
B.4	Predictive Parity	59
B.5	Statistical Parity	60

LIST OF FIGURES

Figure 1.1	Un dimanche après-midi à l'Île de la Grande Jatte, George Seurat	1
Figure 1.2	Research lines Learning Technology & Analytics	3
Figure 1.3	Content and Context of Learning Technology & Analytics . . .	4
Figure 3.1	Bias in an image of ChatGPT 4o	13
Figure 3.2	Identification of sensitive characteristics through cluster analysis	15
Figure 3.3	Division of the dataset into three parts: the training data, the test data, and the validation data	18
Figure 3.4	Application of the model to an average, fictional student	19
Figure 3.5	Application of the model to an average, fictional male/female student	20
Figure 3.6	Ceteris Paribus Plot	20
Figure 3.7	Partial Dependence Plot	21
Figure 3.8	Distribution of the probability of retention after 1 year by highest pre-training	22
Figure 3.9	Fairness analysis by highest level of prior education	23
Figure 3.10	Fairness conclusions per variable	23
Figure 4.1	Fairness conclusions per variable for all Ad and Bachelor's study programmes at The Hague University of Applied Sciences . . .	27
Figure A.1	Division of the dataset into three parts: the training data, the test data, and the validation data	47
Figure A.2	Application of the model to an average, fictional student	48
Figure A.3	Application of the model to an average, fictional male/female student	49
Figure A.4	Ceteris Paribus Plot	49
Figure A.5	Partial Dependence Plot	50
Figure A.6	Distribution of the probability of retention after 1 year by highest pre-training	51
Figure A.7	Fairness analysis by highest level of prior education	52

Figure A.8	Fairness conclusions per variable	52
Figure B.1	Confusion Matrix with 5 ratios for fairness	55

Part I

INTRODUCTION – RESEARCH INTO EQUITY AND FAIRNESS – CONCLUSIONS

*We don't see things as they are,
we see them as we are.*

— Anaïs Nin

1

INTRODUCTION



Figure 1.1: Un dimanche après-midi à l'Île de la Grande Jatte, George Seurat

In the painting 'Un dimanche après-midi à l'Île de la Grande Jatte', we see people spending their Sunday afternoon off on the banks of the Seine. Most people are looking at the river. One girl in a white dress is looking at us – the spectators – while standing next to her mother or babysitter. You can only see the scene by looking from a distance. Even then, the image remains vague. The artist, George Seurat, painted thousands of

dots to form a whole. This technique is called *pointillism*. He worked on it for two years, from 1884 to 1886. The image is precise, but also fleeting: an impression that can quickly fade. Impressionism, Seurat's art style, emerged with the invention of the paint tube in 1841. The paint tube enabled artists to paint outdoors for longer periods. Outdoors, they discovered that reality is constantly changing. They wanted to capture that change in their paintings. Seurat was able to do this like no other.

This painting hangs in Chicago and appears in the 1986 movie 'Ferris Bueller's Day Off'. The movie is about the order and rules of the social system from which Ferris takes a day off school. He drags his friend Cameron along. At this point in the movie – the museum visit – Cameron is lost in the eyes of the girl in the painting. He seems to realize that his life has little meaning if he continues to follow his father's rules.

Today, I will take you on a journey I have been working on for 2 years. An official day off from school, on which we take a step back and try to get an impression of the patterns and rules in our education system. Education is our canvas, the data about our students is the paint, and the technique we use to turn all those dots into a picture is called *learning analytics*.

1.1 LEARNING ANALYTICS, EDUCATIONAL DATA MINING, AND SOCIAL JUSTICE

Learning analytics is "the measurement, collection, analysis and reporting of data about learners and their contexts, for purposes of understanding and optimising learning and the environments in which it occurs" (Siemens & Long, 2011). This definition sounds broad, but it mainly focuses on data arising from the interaction between a student, fellow students, the subject matter, and the teacher.

This is one of the lines of research of the Research Group, but we are researching more (Figure 1.2). Our research also focuses on data at a higher, more systemic level. Think of subjects and subject combinations within study programmes and study careers – I call this *student analytics*. At the level of the educational institution, this is called *institutional analytics*. In the field, these two dimensions are referred to as *Educational Data Mining* (EDM). I also have a line of research that cuts across these different forms of analytics, specifically *inclusion analytics*, in which I investigate equity in education.

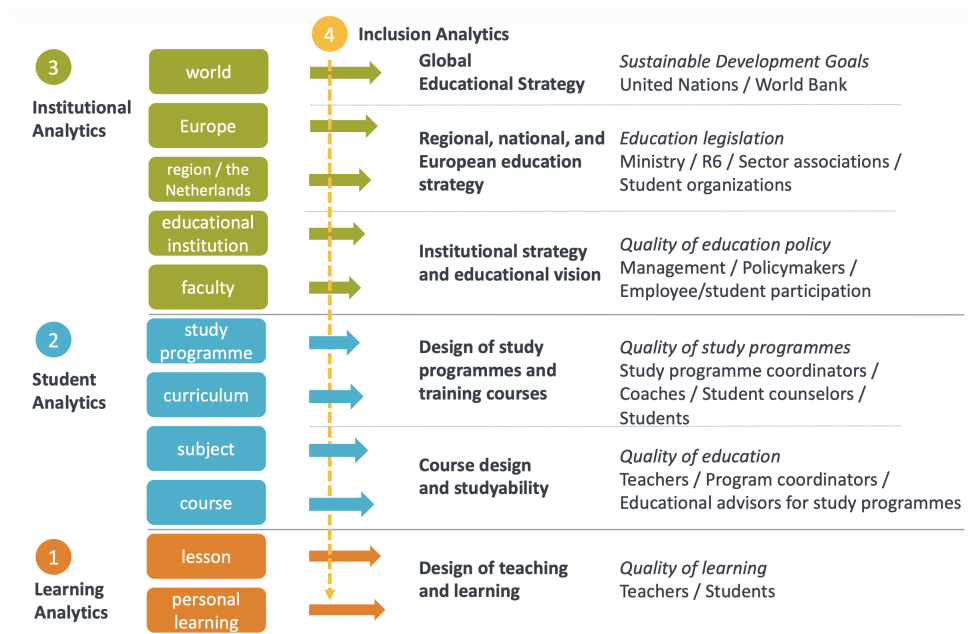


Figure 1.2: Research lines Learning Technology & Analytics

Learning analytics focuses on *the content* of education; EDM focuses on the *context* of education. Because the data comes from different systems, these areas are quite artificially separated from one another. Learning analytics uses more informal data from learning management systems (LMS), while EDM uses formal data from the administrative systems of educational institutions, such as the Student Information System (SIS), Studylink, and the Education Executive Agency (Dienst Uitvoering Onderwijs; DUO) (Figure 1.3).

This practical difference makes it more difficult to understand our education system and to assess its effect on the inclusion and success of students. If learning analytics is applied without context, it can lead to the idea that students must primarily adapt their attitudes or motivations to be successful in education. If the student succeeds in this, it is their own merit. The flip side is that if they do not succeed, it is their own fault. In both cases, it is not the context, the system, that is at fault. This idea, also known as meritocracy, is widespread in Dutch education but, in my opinion, places the responsibility for success largely in the wrong place.

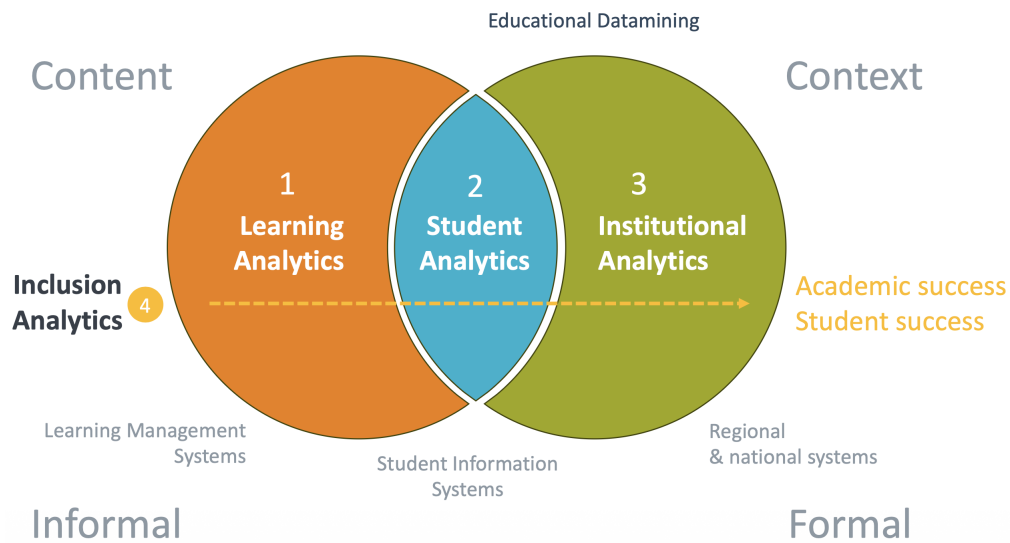


Figure 1.3: Content and Context of Learning Technology & Analytics

So who or what is responsible for success? The terms ‘academic success’ and ‘student success’ refer to success in higher education. The first term – academic success – focuses on the importance of the study programme and would only examine degree completion rates; the second – student success – takes a broader view of student well-being. Personally, I prefer the term academic success. I believe that virtually every student wants to leave higher education with a degree and that most students are happier as a result. A second reason for my preference is that it shifts the focus to the study program and its organization. It is the responsibility of everyone in higher education to guide students as best they can and protect them from problems such as depression or burnout on their path to this goal. A good example of an intervention that fits this is the abolition of the Cum Laude degree for medical students.

If the responsibility lies with the education system, then the question is whether the education we provide is socially just for all students. To answer this question, we must be aware of possible injustices in the system we have built. This is today’s topic: No Fairness without Awareness.

1.2 PURPOSE AND STRUCTURE OF INAUGURAL ADDRESS

In this inaugural address, I will discuss the results of a statistical study on equity. I conducted this study for all bachelor’s and associate degree study programmes at our university of applied sciences for the years 2011 to 2022. I have two goals. First, I want

to improve the position and success of minorities within our university of applied sciences by gaining a better understanding of equity. Secondly, I want to explain how I conducted this study. I hope this will help advance this type of research on equity in Dutch education.

To achieve this, I will first delve deeper into the concept of equity and my personal motivation for this topic. I will then present a research method that allows us to analyze the degree of bias in study data. My aim is to determine a degree of fairness. fairness. I will conclude with conclusions, applications of the insights, risks and recommendations, and plans for further research.

*Where the needs of the world and your talents intersect,
lies your vocation.*

— Aristotle

2

EQUITY IN HIGHER EDUCATION

My research focuses on equity in education. My motivation for this theme is linked to who I am, the family I come from, and how education has enabled me to emancipate myself. My development has been influenced by a quest for equity, both for myself and for my parents. I want to tell this personal story because I believe that virtually every student has a story like this. I try to give these stories a face using learning analytics.

2.1 PERSONAL MOTIVATION FOR EQUITY

I come from a family of educators—both my parents were teachers. Both my father and mother come from single-parent families. My mother's parents divorced in 1944, so she was raised as an only child by her mother. My father, who was slightly older, grew up in a family that was members of the NSB (the Dutch Nazi party), and his father took his own life at the end of the war. They found each other in this shared, complex background of loneliness and social exclusion.

Their teacher training programme was crucial for my parents. Education was their chance to lead a better life and gain social and societal recognition. When I was a little boy of 3, my mother took me to her classroom because I didn't want to stay with the babysitter. My parents instilled in us the importance of education. In our family, I was the first to go to college: a first-generation student. Indirectly, it was also an opportunity for my parents to fulfill their ambitions. For me, it was a chance to understand myself better and ultimately lead a *different* life. I knew from an early age that I was homosexual. Studying and starting a new life in Utrecht allowed me to leave

the Reformed environment in which I grew up behind me and—in a different way—find a place for it. Ultimately, I have ended up here at The Hague University of Applied Sciences today. Now it seems as if I have only myself to thank for that. That is not the case. The fact that I am standing here today as a professor in Learning Technology & Analytics is the result of a series of opportunities that others—many of whom are present here today—have given me. I was able to seize those opportunities because the conditions were favorable for me. Any bottlenecks that existed were therefore manageable for me. That interplay of opportunities, context, and conditions fascinates me.

2.2 THE IMPORTANCE OF HIGHER EDUCATION FOR EQUITY

Unfortunately, the Dutch education system promotes inequality of opportunity. Because students have to choose a level at age 12, differences are magnified at an early age (Copier, 2022). Once in a ‘lower’ school career, it takes more effort to move up to higher education (Elffers, 2022). Access to higher education is seen as the ultimate form of equity. And with good reason.

Higher education institutions play an important role in equity, as emphasized in the United Nations’ Sustainable Development Goals (SDGs). Goal 4.3 states: “By 2030, ensure equal access for all women and men to affordable and quality technical, vocational, and higher education, including university.” (“SDG Nederland - 4 Kwaliteitsonderwijs,” n.d.). Higher education is a driver of personal development and strengthens young people’s position in society (Baum, Ma, & Payea, 2010; Ma, Pender, & Welch, 2016; Perna, 2005; Tinto, 2012). I am not saying that vocational study programmes cannot offer this, but research shows that highly educated people are more satisfied with their work and lives, are more likely to have a job, higher incomes, and live healthier lives (Murray, 2009). Our society rewards jobs for highly educated people more, which reinforces this trend (Belfi et al., 2018).

2.3 LIMITATIONS IN THINKING ABOUT EQUITY AND HIGHER EDUCATION

The earlier definition from the SDGs sounds fair, but has two limitations. This definition only concerns *access* to education and equity for men and women. This is a start, but it is not enough. Equity is not only about equal access, but also about equal outcomes: equal opportunities to obtain a degree and equal opportunities to find further education or

work at the appropriate level (Elffers, 2022). To achieve equal outcomes, we sometimes have to treat students unequally. A second shortcoming is that the definition is limited to equity based on *gender*. There are other characteristics that we need to take into account. I will explain this point in more detail below.

The Netherlands Association of Universities of Applied Sciences (Vereniging Hogescholen; VH) and the Ministry of Education, Culture and Science (Ministerie van Onderwijs, Cultuur en Wetenschap; OCW) also take too narrow a view of equity. In 2019, the VH drew up priorities for improving inclusivity and diversity in higher education (*Professionals voor morgen - Strategische agenda, Vereniging Hogescholen 2019-2023*, 2019). In these priorities, the VH does not use the term 'equal opportunities'. Unfortunately, it seems that the VH reserves this term for the positioning of universities of applied sciences themselves as fully-fledged institutions alongside universities, as evidenced by the 'Position paper on equal opportunities for university of applied sciences students' from 2023 (2023). A limited view of equal opportunities is also reflected in the recent publication by the Ministry of Education, Culture and Science on the measures to reduce the length of studies. The report "Talent in the Right Place" (2024) focuses on the accessibility of higher education, but does account for equal opportunities to obtain a degree. The proposed measure to combat long-term student status will further exacerbate inequality of opportunity. The attractive title of the report, therefore, certainly does not reflect the effect we can expect.

With regard to equity, we can conclude that there are differences in opportunities, but that offering equal resources and guidance does not solve this. Unequally distributed resources and guidance can lead to equal outcomes. However, a fundamental change in the system is needed to achieve social justice. To this end, it is necessary to analyze equity within the system.

*Algorithms don't make things fair.
They repeat our past practices, our patterns.
They automate the status quo.
We have to check them for fairness.
Algorithms can be interrogated.
And they will tell us the truth every time.*

— C. O'Neil, 2017

3

ANALYSIS OF BIAS AND FAIRNESS IN STUDY DATA

Now that we have a better understanding of equity, I would like to discuss the concepts of *fairness* and *bias*¹. I will link these to the inflow, progression, and outflow of students in higher professional education.

3.1 DIVERSITY IN INTAKE AND OUTFLOW

Nothing is as diverse as the intake and outflow in higher professional education. Students enter via vocational education, senior general secondary education, pre-university education, with a foreign degree, or a 21+ test. They enroll in full-time programs, but we also offer part-time and dual programs for students who want to combine work and study. Students from pre-university education often first try a study programme at a university, but then switch to higher professional education. And then there is also the inflow from other study programmes within higher professional education, when students terminate a study programme prematurely. On average, no less than 1 in 3 students at The Hague University of Applied Sciences start a study programme as a switcher ($M = 32.9\%$; internal switch: $M = 8.8\%$; external switch: $M = 24.1\%$).

¹ To avoid confusion, I will use the original English terms in this inaugural address rather than a Dutch translation.

The outflow is just as varied: naturally, some students leave because they have obtained a degree, but a large proportion of students leave their study programmes prematurely, either voluntarily or involuntarily. Students drop out involuntarily because they may have found the program or studying too complex, it was the wrong choice of study, or because they already have a degree, and therefore have a plan B. Voluntary reasons may include continuing to study at a university with a higher professional education propaedeutic degree or changing the type of education. On average, 41% of students drop out of The Hague University of Applied Sciences after one year ($M = 40.8\%$). Of these, an average of 6% do have a propaedeutic diploma ($M = 6.3\%$). footnoteAnd then there is the group of students who deregister before February 1. They do this to avoid receiving a negative BSA. They then re-enroll in the same study programme a year later to complete it. This is not a dropout, but it is not really progression either. On average, this is only 1% ($M = 0.9\%$) of those who re-enroll.

What we see in any case is that students are extremely creative in their *student journey* and thus make their education more flexible. This leads to the question: To what extent is there equity in this intake and progression across different student groups? We will answer this question based by analyzing the study data. But isn't that data biased? The fact that study data may contain bias is precisely what will give us insight into the extent of equity.

3.2 CONCERNS ABOUT EQUITY AND BIAS IN STUDY DATA

In statistical analyses, equity revolves around fairness, bias, and algorithms (Barocas, Hardt, & Narayanan, 2023). *Bias* is "a systematic error in decision-making processes that leads to unfair outcomes" (Ferrara, 2023). If bias has negative effects on people, it can lead to discrimination. An *algorithm* is a formula used to predict a future outcome. *Fairness* means that algorithms must not have discriminatory or unjust consequences (Shin, 2022). A fictional example of an algorithm is:

Gender = Female + Age = 18 years + Registration date = April 7 + Previous study programme = HAVO + Study programme = B Communication & Multimedia Design: Chance of retention after 1 year = 62%, so Retention after 1 year = TRUE; after all, a 62% chance rounds up to 100%.

To illustrate bias, I have an example from when I used ChatGPT-4 with the DALL-E engine. I asked in English to create an image that – based on the height of elementary

school students – could explain a normal distribution ² The image that was produced was problematic on many fronts; for this inaugural address, the problem is the representation of a group of white children in American boarding school outfits (Figure 3.1). Apparently, this is the average image of happy elementary school students.

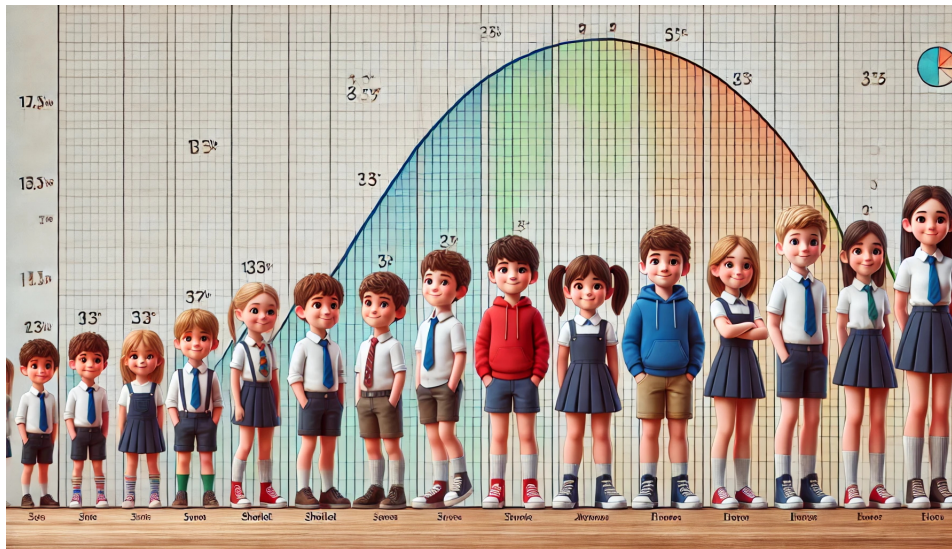


Figure 3.1: Bias in an image of ChatGPT 4o

Concerns have been raised from various quarters about the use of algorithms in education and equity. A well-known example is the algorithm used by DUO to detect potential fraud in student grants for students living away from home (Ersoy & Gaag, 2023). This algorithm flagged students with a non-European migration background as fraudsters with striking frequency. Further investigation revealed that whether a student lived near his or her parents was an important selection criterion in the algorithm. This is more common among students with a migrant background (“Vervolgonderzoek bevestigt indirecte discriminatie controles uitwonendenbeurs,” 2024). Since last week, it has been announced that these students will be compensated for this (Stift, 2024).

The Netherlands Institute for Human Rights has commissioned research to identify risks of discrimination, exclusion, and inequality of opportunity (Smeets, Geurts, &

² Performed in ChatGPT-4o on 10272024. The original prompt was: “Imagine you work at a primary school with 200 children aged 4 to 12. If you ask the children to line up according to height, with shorter children on one side and taller children on the other, you would observe a pattern. Most children would have average heights for their age, so they would cluster in the middle. However, a few children would be much shorter or much taller than the average, standing at the ends of the line. This arrangement creates a curve when you plot the heights on a graph: the highest point is in the middle, where most children are, and it gradually slopes down on both sides, where the few, very short and very tall children stand. This shape is known as a bell curve or normal distribution, which is common in natural measurements like height, weight, and test scores.”

Helvoirt, 2024; “ScienceGuide,” 2024). The Dutch Data Protection Authority (Autoriteit Persoonsgegevens; AP) also sees major challenges regarding equity and the use of algorithms (*Sectorbeeld Onderwijs 2021-2023*, 2024). In particular, there is concern that bias in the study data used for algorithms leads to inequality of opportunity and discrimination. It is striking that no concrete examples were found in practice in this study. Nevertheless, the recommendation is that teachers check for any potential bias in the results. Finally, the European AI Act has identified a number of algorithmic applications in education as high risk. Educational institutions are required to demonstrate that there is no bias in the algorithms they apply or use via software (Engelfriet, 2024).

There is, therefore, growing social concern about possible inequality of opportunity and discrimination in education, but the responsibility for finding a solution is often placed on educational institutions and, in particular, teachers. In my opinion, this is an almost impossible task. This is particularly because we are part of a *system* that promotes inequality. As a result, inequality of opportunity is also present in the study data, in algorithms, in AI, and in all decisions derived from them. We must therefore start with the study data itself to analyze which groups systematically have fewer opportunities, so that we can act on this consciously.

3.3 SENSITIVE CHARACTERISTICS VERSUS SPECIAL PERSONAL DATA

When examining fairness, we look at *sensitive characteristics*, not to be confused with *special personal data*. These are data such as ethnic background and health. Sensitive characteristics are data such as gender identity, disability, age, social background, birthplace, property, language, and political beliefs as defined in the Handbook on European Non-Discrimination Law (*European Union Agency for Fundamental Rights*, 2018). Special categories of personal data are defined in the General Data Protection Regulation (GDPR). These characteristics overlap only partially. Sensitive personal data are characteristics on which discrimination can occur. Special categories of personal data are protected to prevent violations of privacy.

The inclusion of sensitive characteristics in analyses is not without controversy. Even if it is not special personal data, some people believe it is better not to include this data in analyses in order to prevent bias (Pedreshi, Ruggieri, & Turini, 2008). This is a misconception (Hardt, Price, & Srebro, 2016): omitting these characteristics does not mean that they have no influence on the results of an analysis or the decisions that arise from it. They can still be reflected through other variables in a dataset. Suppose we have a dataset containing the date of enrollment in a study, age, and gender of students.

If we analyze this with a cluster analysis, an algorithm based on patterns in age and enrollment date can still see a difference between men and women (Figure 3.2). We want to discover possible bias in existing data. In this way, we can prevent bias in the development of algorithms in higher education from being translated into policy and leading to institutional discrimination. Here too, the principle of “No fairness without awareness” applies. To better understand and promote equity, we should not omit this data; rather, we should analyze it.

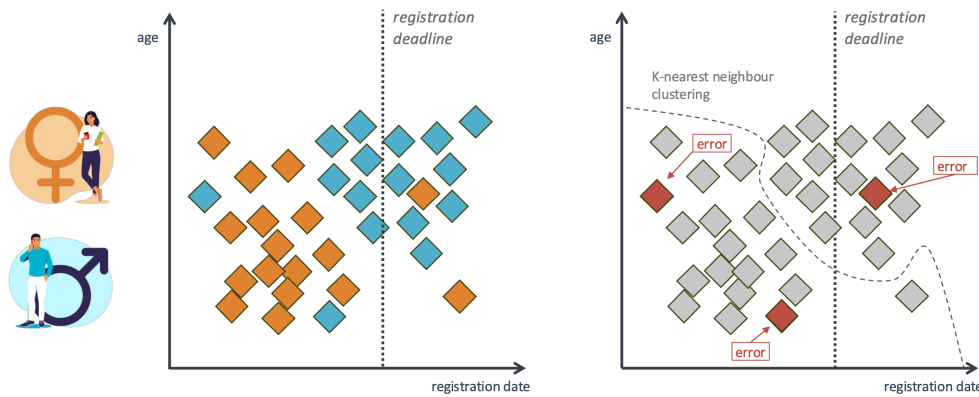


Figure 3.2: Identification of sensitive characteristics through cluster analysis

3.4 CURRENT RESEARCH INTO FAIRNESS IN (HIGHER) EDUCATION

There is one important caveat, however. Research on fairness and bias primarily originates in the United States (US). The datasets that were important in the development of this type of research were datasets for granting loans, bail after arrest, jobs, or – relevant to our field of work – places in university study programmes (Dwork et al., 2011). The main question in these studies is whether ethnic minorities were treated equally in these procedures. Curiously, this introduces a bias into the research itself. As we saw earlier, the American context is not necessarily representative of the European or Dutch context. There are biases toward sensitive characteristics that are formally recognized nationally or regionally, such as “veteran” or “ethnicity” in the US. The most researched characteristics in fairness in higher education are ethnicity, nationality, and gender (Baker & Hawn, 2022). However, other relevant characteristics, such as language, disability, urban or rural origin (*urbanicity*), socioeconomic status, and prior education, are underrepresented.

3.5 FAIRNESS ANALYSIS

A range of statistical techniques is now available for developing prediction models. We create prediction models based on historical data. Suppose we have 10 years of study data on the success of first-year students in a study programme, with variables such as the gender of students, their age on October 1, and their average final exam score in their previous education. We can use this to predict an outcome variable, namely whether first-year students will continue their studies after their first year: retention after 1 year. We then apply this prediction model to *new* first-year students. So we look back to look ahead; this can offer us new insights – we go from *hindsight* to *foresight* and ultimately *insight*.

The process of arriving at a prediction model for retention after 1 year and thus calculating fairness consists of three steps: 1) developing prediction models³, 2) determining the importance of variables, and 3) calculating fairness⁴.

3.5.1 Step 1 - Developing prediction models

The first step is to prepare the data and build a number of predictive models to predict retention after 1 year per program.

- After completing the necessary procedures, we first collect historical study data. For each first-year student in each study program and academic year—the cohort—there is a row in the dataset containing study data. We combine data where necessary and resolve any errors.
- We remove rows without an outcome variable, retention after 1 year,⁵.
- We then resolve rows with missing data, as prediction models cannot handle these. It is important to first investigate missing values thoroughly to understand why they are missing. For example, vocational students and students with foreign prior education do not have an average grade from their prior education. We fill in rows with missing data (imputation). For each variable we fill in, we create a new variable regardless of whether there was an original value or not

³ For this research, we used the *tidymodels* package in R (Kuhn & Wickham, 2020).

⁴ For the research into variables and fairness, we used the *DALEX*, *DALEXtra*, and *fairmodels* packages in R (Biecek, 2018; Maksymiuk, Gosiewska, & Biecek, 2020; Wiśniewski & Biecek, 2022).

⁵ This occurred in our research for a study programme that had only just started.

(VariableName_Missing). If too many missing values are missing (50% or more), we remove that variable from the dataset.

- Next, we examine variables that are strongly correlated with each other. Examples include the average grade for the central final exam and the average grade for the school exam. For strongly correlated variables, we normally retain the most predictive value, but in this study, we retain all variables so we can understand their relationship afterwards.
- Next, we divide the data into three parts: the training data, the test data, and the validation data. For example, we split the total dataset into 60% for training, 20% for testing, and 20% for validating the model we have developed (Figure A.1). In this division, we ensure that the ratio of students continue their studies after 1 year is the same everywhere (stratification). How well the model performs on the validation data will determine its quality, such as its accuracy.
- Now we create the prediction model. It is important to create a prediction model for each study programme, because the students who enroll in each study programme are unique. Enrollment depends on the type of study programme (associate degree or bachelor's degree), the form of the study programme (full-time, part-time, or dual), the nature of the study programme (the subject), and, for the geographical distribution of students, the degree of competition in the region. For each study programme, we fit two prediction models for retention after one year: a logistic regression and a random forest. We choose these models because they make it easy to explain which variables are included and what each variable contributes to the prediction.
- Finally, we calculate the quality for each study programme and each model using the Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC). The AUC/ROC measures the extent to which a prediction model correctly classifies positive outcomes as positive and negative outcomes as negative. In our example, this means that the model accurately predicts which students will or will not continue their studies after 1 year. The model with the highest Area Under the Curve (AUC/ROC) forms the basis for the next step: calculating fairness.

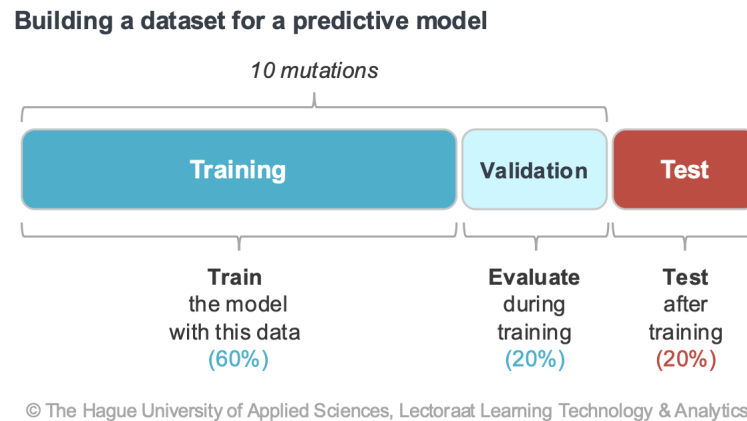


Figure 3.3: Division of the dataset into three parts: the training data, the test data, and the validation data

3.5.2 Step 2 - Determining the importance of variables

The second part of the analysis consists of gaining a better understanding of the variables in the prediction model:

- After developing two models for each study programme, we continue with the model that has the highest degree of accuracy. For that model, we identify which variables are important for predicting retention after 1 year and how they contribute to the probability of retention.
- We then apply the models to the average student in a study programme. We calculate the average student by averaging the entire student population within the study programme for continuous variables, and by using the most common value for categorical variables in order of importance within the model. See Figure A.2.
- We then repeat the analysis for different groups. We examine gender, highest prior education, and type of connection⁶. Here, too, we calculate an average student for each group and then apply the prediction model. See ??.
- To better understand the patterns we find in this, we apply follow-up analyses to assess the ranking of variables, better understand the individual effects of

⁶ Type of connection refers to how the student entered the program after completing their highest prior education: was it directly after prior education, after a gap year, after an internal or external switch, after a colloquium doctum or 21+ test, as a second degree, or is the connection unknown?

variables, and estimate the degree of variance within groups: a Shapley analysis, a Ceteris Paribus analysis (Figure A.4), and a Partial Dependency analysis (Figure A.5).

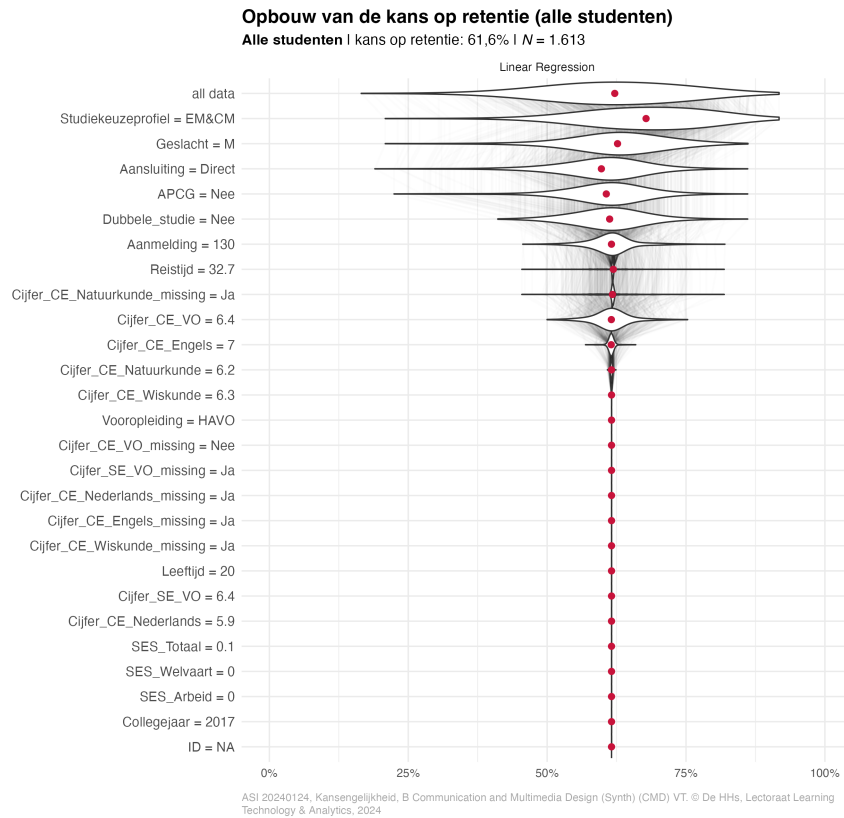
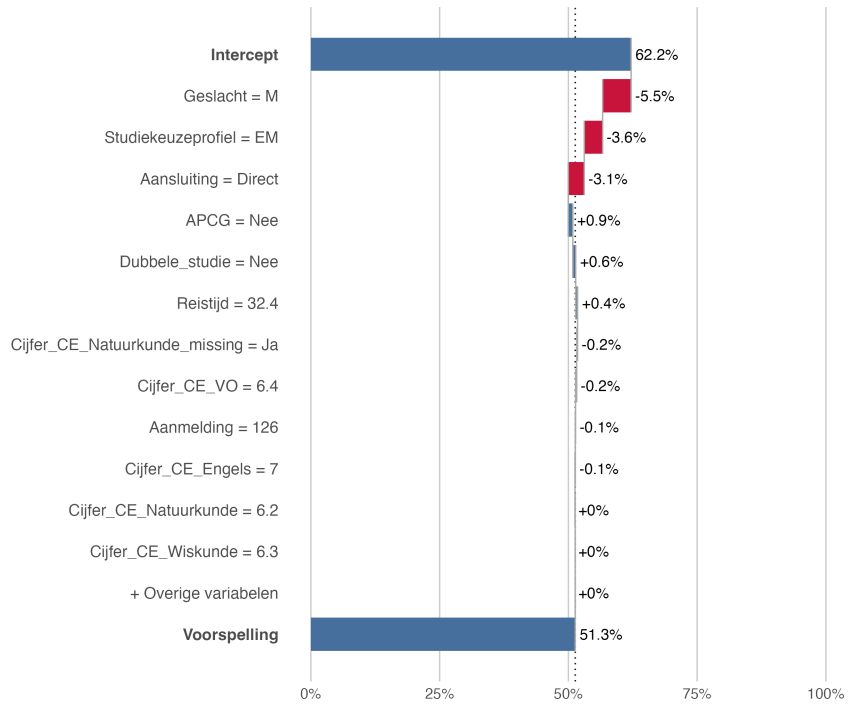
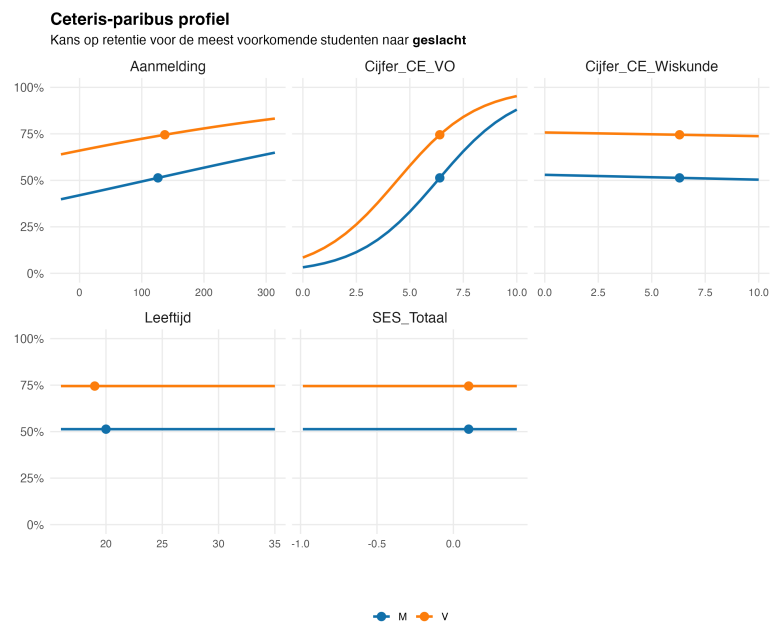


Figure 3.4: Application of the model to an average, fictional student

Opbouw van de kans op retentie naar geslachtGeslacht: **M** | kans op retentie: 51,3% | $N = 915$ van 1.613 (56,7%)

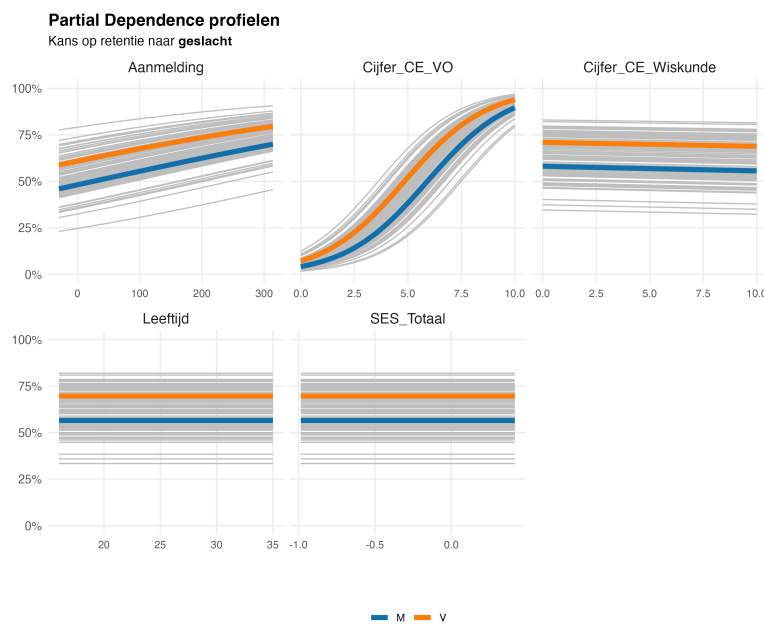
ASI 20240124, Kansengelijkheid, B Communication and Multimedia Design (Synth) (CMD) VT. © De HHs, Lectoraat Learning Technology & Analytics, 2024

Figure 3.5: Application of the model to an average, fictional male/female student



ASI 20240124, Kansengelijkheid, B Communication and Multimedia Design (Synth) (CMD) VT. © De HHs, Lectoraat Learning Technology & Analytics, 2024

Figure 3.6: Ceteris Paribus Plot



ASI 20240124, Kansengelijkheid, B Communication and Multimedia Design (Synth) (CMD) VT. © De HHs, Lectoraat Learning Technology & Analytics, 2024

Figure 3.7: Partial Dependence Plot

3.5.3 Step 3 - Calculating fairness

The third and final part of the process is calculating fairness:

- First, we analyze the probability distribution within groups; this gives us a good picture of the distribution of probabilities in the prediction model for each group and allows us to compare them effectively. See Figure A.6.
- Next, we apply the 4/5 criterion to a number of performance values, including accuracy⁷. As the privileged group, we use the most common subgroup within a variable⁸. If two or more values fall outside the bandwidth of the 4/5 criterion, there is a lack of fairness for the protected group in relation to the privileged group. This can be negative, but also positive for some groups. We also calculate the 'total loss' per variable. The larger this number, the greater the differences within a study programme. See Figure A.7.

⁷ A total of 5 ratios were assessed: Accuracy Equality, Equal Opportunity, Predictive Equality, Predictive Parity, and Statistical Parity. See Appendix A for an explanation of these ratios.

⁸ Suppose there are more men than women in a study programme, then 'Men' is the privileged group; in the event of a tie, one of the two is chosen.

- Finally, we summarize the findings in a table to see where there is bias and for which groups. We distinguish between groups for which we have sufficient data to make a statement and those for which we do not⁹. See Figure A.8.

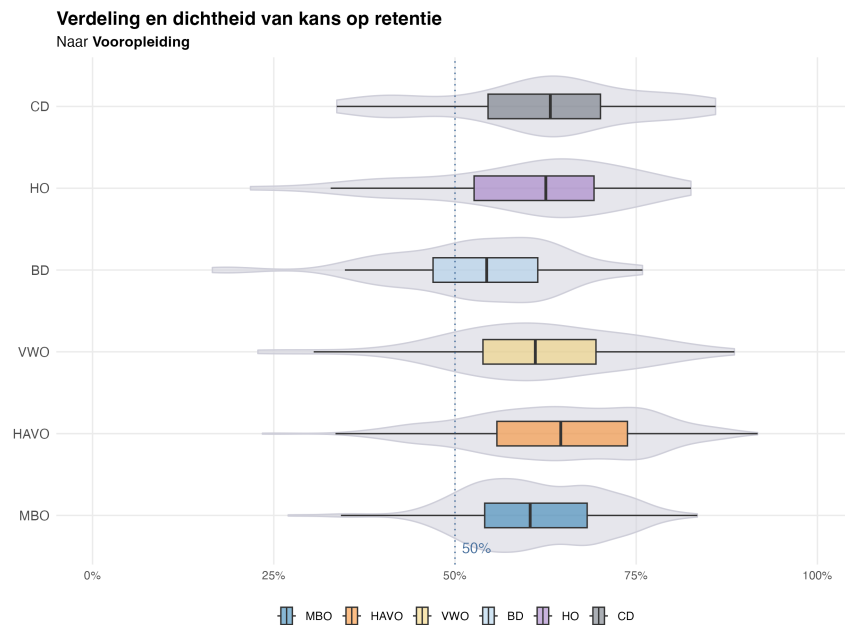


Figure 3.8: Distribution of the probability of retention after 1 year by highest pre-training

⁹ NB. It may be that the privileged group is the group that suffers from bias. This can be determined if all other groups have a negative or positive bias

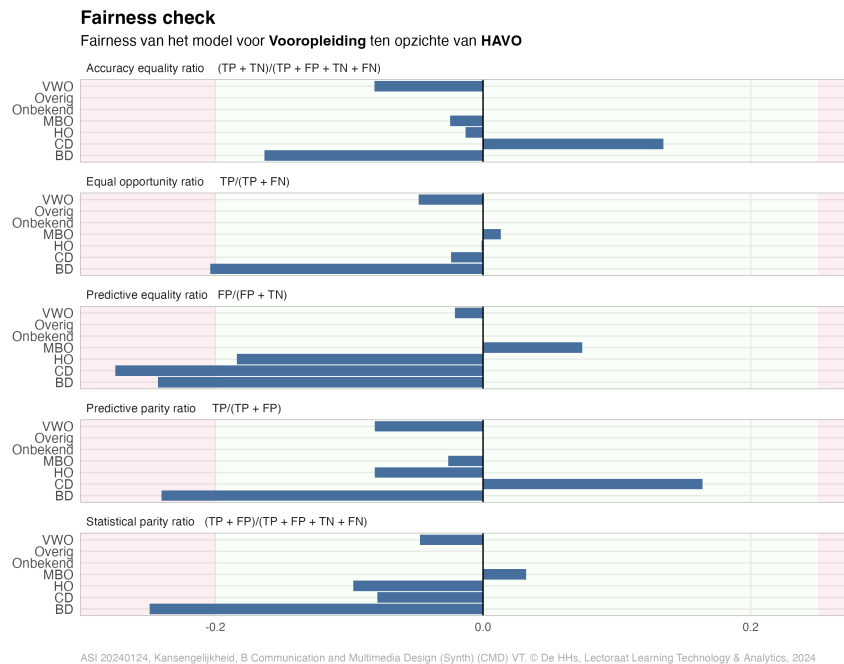


Figure 3.9: Fairness analysis by highest level of prior education

Variabele	Groep	Bias	Geen Bias	Negatieve Bias	Positieve Bias
<i>Geslacht</i>	V	Nee	5	0	0
<i>Vooropleiding</i>	BD	Ja	1	3	1
	CD	Ja	0	0	5
	HO	Nee	5	0	0
	MBO	Ja	2	3	0
	VWO	Ja	3	0	2
<i>Aansluiting</i>	2e Studie	Ja	2	2	1
	Na CD	Ja	0	0	5
	Switch extern	Nee	4	0	1
	Switch intern	Nee	4	1	0
	Tussenjaar	Ja	2	0	3

Figure 3.10: Fairness conclusions per variable

We have completed this procedure for all associate degree and bachelor's study programmes at The Hague University of Applied Sciences, in all variants. For each study programme, we have incorporated the procedure into a self-explanatory website

with a separate page for each step, including the underlying code. Starting today, the Research Group will make the source code for this method available to anyone in education who wants to apply this approach. Further dissemination will take place via the Npuls program in collaboration with The Hague University of Applied Sciences.

*In God we trust.
All others must bring data.*
— W. Edwards Demming

4

CONCLUSIONS, RECOMMENDATIONS, AND FOLLOW-UP RESEARCH

Now that we have gone through all the steps to analyze fairness, I will discuss the most important findings, the possible applicability of this approach, any shortcomings and recommendations, as well as the follow-up research we plan to conduct within the Research Group.

4.1 THE MAIN FINDINGS OF THE FAIRNESS STUDY

- **Fairness can be analyzed and discussed** – The first finding is that this approach appears to be a repeatable and scalable research method for determining bias in study data. The research can be carried out for all study programmes at our university of applied sciences, and fairness can be determined using fairly basic data. Initial discussions with colleagues from various study programmes show that the approach is easy to explain and provides immediate insight into groups of students who are disproportionately less likely to progress after their first year of study.
- **There is still much to be gained from improving fairness** – The second finding is that most of the 66 study programmes show a lack of fairness for some groups of students (see Figure 4.1). Only six study programmes were exceptions to this, one of which was a selective study programme. It is also striking that only one study programme shows a negative bias toward retention after one year among women, whereas 20 study programmes show a negative bias toward men. In 20

study programmes, there is still much to be gained in terms of equity for MBO students, while in 29 study programmes, VWO students have disproportionately more opportunities.

- **Sensitive characteristics are expressed in non-sensitive characteristics** – A third finding is that some variables appear unimportant, but express an underlying characteristic. The variable average Central Examination score, for example, is relevant because it is missing for MBO students and students with a foreign prior education. This makes this variable more valuable in prediction models than the underlying variable, the highest prior education. Variables such as highest prior education, secondary school grades, and enrollment date are often proxies for socioeconomic status and possibly migration background. They appear to reflect abilities or motivation, but are influenced by underlying socioeconomic factors. More research into this is needed.
- **There is potential for using these insights, but little demand yet** – A fourth outcome concerns the use of these analyses. Although THUAS' institutional plan focuses on removing study bottlenecks "Onderzoekend leren met impact," 2022, requests to investigate equity have not yet been frequent among the study programmes studied. Only within the Economics sector have agreements been made to assess equity (*Taakopdracht sectorale verkenning hoger economisch onderwijs (HEO) 2020-2021, 2020*). Requests usually focus on increasing enrollment or reducing dropout rates. The good news is that improving equity can increase enrollment and reduce dropout rates. Here too, *No fairness without awareness* applies.

- **A tool for investigating equity within THUAS and the region** – This research provides THUAS with a tool to determine whether there are groups of students who have disproportionately fewer opportunities in their studies, and to then take action. The research conducted offers the opportunity to investigate this within the region. This fits in perfectly with the agenda of the Equal Opportunities Alliance of the municipality of The Hague (2019-2021) (*GKA-Agenda gemeente Den Haag, 2019*). “All children and young people have the right to equal opportunities in education. It is important that the talents of all children and young people are fully utilized. This is not only of great importance for the future of children and young people themselves, but also for the future of our society” The first areas we want to test regionally are the progression from secondary vocational education to higher professional education and internship discrimination.
- **A national, statistical research method for examining equity in further education** – In addition, this research offers all educational institutions a tool for quantitatively examining equity. I am pleased to announce that we will make this method available as open access and open source. We have found Npuls willing to adopt the method’s upscaling and underlying source code through the Center for Educational Data Analytics.
- **A synthetic study dataset that students can use to conduct research** – Finally, we have developed a synthetic dataset based on existing study programmes at The Hague University of Applied Sciences to explain the method. This dataset has the characteristics of all study programmes, including the distribution of variables, but is completely fictional. Students at The Hague University of Applied Sciences can use this set to conduct their own research with the Research Group.

Possible shortcomings, risks, and recommendations

Research into algorithmic fairness in higher education shows that field is evolving from unknown to known bias (Baker & Hawn, 2022). Our research contributes further to this. However, like any research, our research also has shortcomings and risks. I will discuss these and also provide recommendations to counteract them.

- **The approach does not take individual differences into account** – In the structure of fairness analysis, we do examine the degree of variance within groups, but we do not take individual fairness into account. It is possible that the statements we make about groups do not apply to individuals within those groups, because other, unknown characteristics may work to their advantage or disadvantage.

My recommendation is to refine the method using transcendent analyses, for example, of combinations of prior education, gender, and intake type. Furthermore, always take the student's personal situation into account.

- **The approach does not take special personal data into account** – Adding special personal data, such as data on migration background or support needs, can give fairness analyses even more context. My recommendation is to include this data in this type of research. In the current study, I deliberately excluded this data from the scope. Not because I do not consider it important—quite the contrary—but because it is new for our university to collect and combine data on a large scale for this type of research. Another factor is that, because informed consent is a prerequisite for the processing of special personal data, we are introducing a new type of bias: consent bias. In particular, students who have greater confidence in an educational institution are more inclined to consent to the collection of this type of data.
- **The research does not yet provide any tools for resolving a possible lack of equal opportunities** – This method does identify the problem, but does not indicate where a solution might lie. To be clear, this is not the adjustment of the prediction model or the study data. Because this method provides insight into fairness per group, solutions must be sought in possible interventions that could benefit specific groups of students. Teachers, supervisors, and administrators are bound by a decision space. Alternative options are excluded or difficult to achieve, such as postponing or abolishing the BSA, alternative testing, exemptions, extra time, etc. During the coronavirus pandemic, these options were available, which makes them interesting to study. Research into differences in study success between students before and after the coronavirus pandemic (Vooren et al., 2024) shows that abolishing the binding study advice would already contribute to equity in retention and diploma success. What would that broader decision space look like if other measures were also applied to all students, such as extra time for exams, a class absence card, different types of tests, or an alternative interpretation of certain learning outcomes? My recommendation is to consult further with teachers and students to determine whether there is a lack of equity that the study programme can and wants to address. I advocate for group arrangements rather than the individual application of prediction models. First of all, it must be determined to what extent a solution can be found in education. This can be done by further examining individual subjects in the curriculum. Perhaps this will

reveal that certain skills are insufficiently present. In response to this, targeted remedial education could be offered to the entire student population, so that students can participate voluntarily. If, for example, panel interviews reveal that barriers lie in the more personal sphere of life, solutions may be sought within the care structure. A difference with current practice is that solutions are not only sought after problems have been identified by examination boards, supervisors, or psychologists, but are offered proactively.

- **A lack of equal opportunities is not the same as discrimination** – There is a risk that we confuse the outcome of equal opportunities with discrimination. The aim of this study was to investigate equal opportunities in the context of higher education and The Hague University of Applied Sciences in particular; this can be separate from discrimination. Discrimination is “treating people differently, disadvantaging them, or excluding them on the basis of personal characteristics” (“Wat is discriminatie? | Mensenrechten voor jou | College voor de Rechten van de Mens,” n.d.). The question is whether any disadvantage is caused *deliberately*. I do not expect this to be the case. My recommendation is to investigate whether students experience the current education system and associated procedures as discriminatory. Within the Center of Expertise Global & Inclusive Learning, there are colleagues specialized in this type of research.
- **These insights can be used to remove bottlenecks, but also to reduce them and create additional barriers** – There could be a tendency to use insights from this research to discourage students who have less chance of success in the current education system. This could, in principle, contribute to what some see as the quality of the study programme. Unlike algorithms, people do have an ethical compass, which I strongly appeal to here. I recommend that this appeal be accompanied by a code of conduct on privacy, ethics, and AI for our university. This is in line with the AI Act, which came into force on August 1 of this year and characterizes education as a high-risk area subject to additional safeguards. One of these safeguards is that educational institutions must demonstrate that any algorithms they use take bias into account and that they train their staff to recognize and counteract bias.
- **These predictive models may not be used for individual application to current or prospective students** – In line with the previous risk, there may be a desire to use these predictive models for individual application. Even if this were done with

the student's *informed consent*, it is still not permitted. They have been developed solely to determine the degree of fairness.

On this point, my recommendation is that, for the development of predictive models in education or guidance, collaboration with the Research Group should be sought to take into account any bias and, together with students, to further develop new instruments.

4.3 FOLLOW-UP RESEARCH

I would like to conclude with a follow-up research plan to broaden and deepen our approach.

- **Other success variables** — Other success variables, such as the likelihood of obtaining a degree. It is almost certain that the new cabinet's announced measure to shorten the length of study will have a negative impact on educational equity. Students who need more time to reach higher education are also likely to need more time to complete it.
- **Special personal data** – We also want to investigate the impact of including special personal data with informed consent.
- **Curriculum components** – In addition, we will also investigate equity within the curriculum of study programmes: individual courses, groups of courses, and larger study components. This includes, in particular, possible internship discrimination.
- **Labor market success** — We are also in talks with the Randstad 6 universities of applied sciences to see if we can conduct an equal opportunities study among all students who have studied at our universities of applied sciences, with or without completed study programmes. To this end, we plan to conduct research using CBS microdata.
- **Replication** – We also hope that our research can be replicated at other educational institutions. The upscaling from the Npuls program will certainly help in this regard.
- **Learning Technology** – Finally, within *learning technology*, we have started two lines of research into the effects of interventions, particularly in the use of EdTech.

In addition, we are researching *honesty* in online testing. With the arrival of a new senior researcher on this topic, we expect to make excellent progress in this area.

This brings me to the end of my inaugural address. I have given you an impression of my Research Group and the potential of this type of research. I hope this will enable me to make a significant contribution to learning analytics in the Netherlands and to equity and social justice in the Dutch education system for all our students.

Let's Change. You. Us. The World.

SUMMARY

*Sometimes it is the people no one imagines anything of
who do the things that no one can imagine.*

— A. Turing

In his inaugural address, *No Fairness without Awareness – Applied Research on Equity in Higher Education*, Theo C. Bakker, PhD, introduces the topic of equity in higher education and the role of learning analytics in promoting social justice. From his professorship in Learning Technology & Analytics at The Hague University of Applied Sciences (THUAS), Bakker has been conducting statistical research on equity in higher vocational education (hbo), using student data and AI predictive models to examine the extent to which different student groups are treated equally. He emphasizes that equity of opportunity is about access to education and fair opportunities for study success and advancement into the labor market.

IMPORTANCE OF LEARNING ANALYTICS FOR EQUITY

Learning analytics involves measuring, collecting, and analyzing data about students and their learning environments to optimize instruction and student support. Within this field, Bakker further distinguishes between different types of analytics: *learning analytics*, which examines the interactions between students, teachers and the subject matter; *student analytics*, which examines individual student data; *institutional analytics*, which looks at organizational-level data; and *inclusion analytics*, which focuses on social justice and equity. In the latter type, he examines whether certain groups of students are systematically disadvantaged, for instance, in dropping out.

To that end, Bakker discusses the problem of bias in educational data. Bias is a systematic error in data that can lead to unfair outcomes. Algorithms often reinforce existing biases in the data, leading to educational inequity. He emphasizes that equity can only be realized if we are aware of possible injustices in the system: *No Fairness without Awareness*.

METHOD: ANALYSIS OF BIAS AND FAIRNESS

For his research, Bakker uses machine learning algorithms that predict retention after the first year of study. By analyzing these predictions, he examines whether the odds are fairly distributed between groups, for example, between men and women or between mbo-, havo-, and vwo-students. An essential measure in the study is the 4/5 criterion, which assesses whether the odds for the 'protected group' are at least 80% of those of the 'advantaged group'. If the ratio is smaller, there is a lack of fairness.

Bakker describes three steps in the analysis:

1. *Prepare data and develop prediction models*: The data are cleaned and split into a training, testing, and validation set. Models such as logistic regression and random forest are used to predict which students are likely to pass the first year.
2. *Determining the importance of variables*: The influence of characteristics such as prior education and age on outcomes is examined. A fictitious 'average student' is modelled for different groups to understand differences in odds.
3. *Calculating Fairness*: The 4/5 criterion is used to compare differences in fairness between groups. This criterion is applied to various metrics from prediction models, such as accuracy, equality of opportunity, and statistical equality. The analysis shows where bias occurs and how it affects equity.

RESULTS AND CONCLUSIONS

Bakker's research shows that inequity exists in the majority of programs. At THUAS, for example, women in only one program show a negative bias, while men in 20 programs have a lower chance of retention. Vwo students are more likely to have a positive bias, while mbo students are at a disadvantage. The approach provides concrete insights for educational institutions to identify and analyze inequities. This approach can contribute to interventions specifically aimed at supporting groups of students with a disproportionately higher risk of dropping out. Moreover, the methodology can be scaled up and provides a replicable approach to detect bias in student data. A national-scale-up will be funded by the Npuls Growth Fund Program.

RECOMMENDATIONS AND FOLLOW-UP RESEARCH

Bakker makes several recommendations to promote equity in higher education. First, he advocates using cross-cutting analyses to improve understanding of more complex inequality patterns. In addition, he recommends responsibly including sensitive characteristics, such as migration background and socioeconomic status, in analyses to provide a fuller picture of inequalities. He warns of *consent bias*: students who trust the educational institution are more likely to consent to the use of their personal data, potentially affecting the representativeness of the data.

Another recommendation concerns developing interventions to reduce bottlenecks. Bakker suggests that program managers and teachers in educational institutions should be given greater *decision space*, with more opportunities for flexible, proactive support for students. He refers to measures during the Covid crisis, such as the postponement of the binding study advice, which have shown that policy flexibility increases the chances of study success. Finally, Bakker argues that discrimination and inequity are different phenomena. Not every inequality in outcomes results from discrimination; it can also stem from systemic failures in education.

FUTURE PROSPECT

Bakker's research provides policymakers, educators, and administrators in higher education with valuable insights. His recommendations guide new studies and policies aimed at reducing inequity. In follow-up research, he plans to look at retention, degree attainment, and internship discrimination. The professorship will make the methodology open access available through Npuls, allowing other institutions to adopt and broaden the approach.

Bakker concludes his inaugural address by calling for data-informed decision-making to build a more socially just educational system that gives students a fair chance to succeed.

Let's Change. You. Us. The World.

ACKNOWLEDGEMENTS

*Learn from yesterday, live for today, hope for tomorrow.
The important thing is not to stop questioning.*

— A. Einstein

I would like to express my sincere gratitude to the Executive Board of The Hague University of Applied Sciences, the management of the IT & Design faculty, and the lecturers at the Centre of Expertise Global & Inclusive Learning for the opportunity they have given me to become professor in Learning Technology & Analytics. I hope that I have been able to live up to their expectations. I would also like to thank my former colleagues at the Education, Knowledge & Communication (Onderwijs, Kennis & Communicatie; OKC) department who took over my duties after I left my position as director.

Within the university, I would also like to thank all my colleagues who made this research possible: my colleagues in Legal Affairs for their cooperation in the DPIA for this research, my colleagues in the university library for their support in data management, the members of the ethics committee for reviewing the research proposal, and my colleagues in OCK, the IR & Analytics for providing the research data. Outside the university, I would like to thank my colleagues at Npuls for giving me the opportunity to scale up this research nationally.

Many thanks also to all members of the Learning Technology & Analytics Research Group for their enthusiasm for our research into Learning Analytics of Learning Technology. I learn from you every day. In addition, many thanks to my wonderful management assistant, Titia Dijkstra. Without you, the Research Group would be nowhere! Dear Manika, Thank you for joining my research group. It is a delight working with you.

Finally, many thanks for all the support I have received at home from my dear husband in taking this step. Dear George, when we met, I was an entrepreneur and you were a researcher, but now we have switched places. I would like to thank my parents for instilling in me a love of education and studying.

REFERENCES

- Baker, R. S., & Hawn, A. (2022). Algorithmic Bias in Education. *International Journal of Artificial Intelligence in Education*, 32(4), 1052–1092. <https://doi.org/10.1007/s40593-021-00285-9>
- Barocas, S., Hardt, M., & Narayanan, A. (2023, December). *Fairness and Machine Learning: Limitations and Opportunities*. fairmlbook.org. Retrieved June 22, 2024, from <http://www.fairmlbook.org>
- Baum, S., Ma, J., & Payea, K. (2010). *Education Pays 2010: The Benefits of Higher Education for Individuals and Society* (tech. rep.). CollegeBoard. <https://files.eric.ed.gov/fulltext/ED526357.pdf>
- Belfi, B., Allen, J., Jacobs, B., & Oosterhout, K. V. (2018). *Arbeidsmarktongelijkheid onder hbo'ers: Hoe individuele achtergrondkenmerken van hbo'ers samenhangen met arbeidsmarktsucces op de korte en middellange termijn* (tech. rep.). Retrieved November 21, 2023, from https://www.hbomonitor.nl/application/files/1915/4503/3607/ROA%5C_R%5C_2018%5C_9.pdf
- Biecek, P. (2018). Dalex: Explainers for complex predictive models in r. *Journal of Machine Learning Research*, 19(84), 1–5. <https://jmlr.org/papers/v19/18-416.html>
- Copier, J. (2022). *Tussen idealen en dwalingen. Verhalen over onderwijs*. Garant.
- Dwork, C., Hardt, M., Pitassi, T., Reingold, O., & Zemel, R. (2011). Fairness Through Awareness. *arXiv*. <https://doi.org/10.48550/arxiv.1104.3913>
- Elffers, L. (2022, May). *Onderwijs maakt het verschil - kansengelijkheid in het Nederlandse onderwijs*. Walburg Pers B.V.
- Engelfriet, A. (2024, January). *AI and algorithms Mastering legal and ethical compliance*. Ius Mentis (ICTRecht).
- Ersoy, S., & Gaag, S. V. d. (2023, June). Studenten met migratieachtergrond opvallend vaak beschuldigd van fraude, minister wil systeem grondig nagaan. Retrieved June 21, 2023, from <https://nos.nl/op3/artikel/2479700-studenten-met-migratieachtergrond-opvallend-vaak-beschuldigd-van-fraude-minister-wil-systeem-grondig-nagaan>

- Ferrara, E. (2023). Fairness and Bias in Artificial Intelligence: A Brief Survey of Sources, Impacts, and Mitigation Strategies. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.4615421>
- GKA-Agenda gemeente Den Haag (tech. rep.). (2019, July). Gemeente Den Haag. Retrieved August 1, 2024, from <https://www.gelijke-kansen.nl/gemeenten/den-haag>
- Handboek over het Europese non-discriminatierecht, Editie 2018*. (2018). https://fra.europa.eu/sites/default/files/fra%5C_uploads/fra-2018-handbook-non-discrimination-law-2018%5C_nl.pdf
- Hardt, M., Price, E., & Srebro, N. (2016). Equality of Opportunity in Supervised Learning. *arXiv*. <https://doi.org/10.48550/arxiv.1610.02413>
- Kuhn, M., & Wickham, H. (2020). *Tidymodels: A collection of packages for modeling and machine learning using tidyverse principles*. <https://www.tidymodels.org>
- Ma, J., Pender, M., & Welch, M. (2016). *Education Pays 2016: The Benefits of Higher Education for Individuals and Society*. (tech. rep.). CollegeBoard. <https://files.eric.ed.gov/fulltext/ED572548.pdf>
- Maksymiuk, S., Gosiewska, A., & Biecek, P. (2020). Landscape of r packages for explainable artificial intelligence [Pages 6, 7, 11, 15]. *arXiv*. <https://arxiv.org/abs/2009.13248>
- Murray, J. (2009). The Wider Social Benefits of Higher Education: What do We Know about Them? *Australian Journal of Education*, 53(3), 230–244. <https://doi.org/10.1177/000494410905300303>
- Onderzoekend leren met impact. (2022). <https://www.dehaagsehogeschool.nl/sites/hhs/files/documents/OverDeHaagse-Organisatie-FeitenDocumenten-instellingsplan-02-15-def.pdf>
- O’Neil, C. (2017, April). The era of blind faith in big data must end. Retrieved July 29, 2024, from [The%20era%20of%20blind%20faith%20in%20big%20data%20must%20end](https://www.theeraofblindfaithinbigdata.com)
- Pedreshi, D., Ruggieri, S., & Turini, F. (2008). Discrimination-aware data mining. *Proceedings of the 14th ACM SIGKDD international conference on Knowledge discovery and data mining*. <https://doi.org/10.1145/1401890.1401959>
- Perna, L. W. (2005). The Benefits of Higher Education: Sex, Racial/Ethnic, and Socioeconomic Group Differences. *The Review of Higher Education*, 29(1), 23–52. <https://doi.org/10.1353/rhe.2005.0073>
- Position paper Gelijke kansen voor de hogeschoolstudent* (tech. rep.). (2023, February). Vereniging Hogescholen.

- Professionals voor morgen - Strategische agenda, Vereniging Hogescholen 2019-2023* (tech. rep.). (2019, September). Vereniging Hogescholen. Den Haag. Retrieved July 27, 2024, from https://www.vereniginghogescholen.nl/system/knowledge%5C_base/attachments/files/000/001/067/original/Professionals%5C_voor%5C_morgen%5C_- %5C_strategische%5C_agenda%5C_Vereniging%5C_Hogescholen%5C_2019-2023.pdf
- SDG Nederland - 4 Kwaliteitsonderwijs. (n.d.). Retrieved July 27, 2024, from <https://www.sdgnerland.nl/SDG/4-kwaliteitsonderwijs/>
- Sectorbeeld Onderwijs 2021-2023* (tech. rep.). (2024, January). Autoriteit Persoonsgegevens. Retrieved August 1, 2024, from <https://www.autoriteitpersoonsgegevens.nl/documenten/sectorbeeld-onderwijs-2021-2023>
- Shin, D. (2022). How do people judge the credibility of algorithmic sources? *AI & SOCIETY*, 37(1), 81–96. <https://doi.org/10.1007/s00146-021-01158-4>
- Siemens, G., & Long, P. (2011). Penetrating the Fog: Analytics in Learning and Education. *EDUCAUSE Review*, (46), 31–40.
- Smeets, E., Geurts, R., & Helvoirt, D. v. (2024). *Algoritmen in het onderwijs. Een onderzoek in opdracht van het College voor de Rechten van de Mens* (tech. rep.). KBA Nijmegen / ResearchNed.
- Stift, R. (2024, November). Tienduizend studenten krijgen compensatie wegens discriminatie DUO. Retrieved November 11, 2024, from <https://www.nrc.nl/nieuws/2024/11/11/tienduizend-studenten-krijgen-compensatie-wegens-discriminatie-duo-a4872677>
- Taakopdracht sectorale verkenning hoger economisch onderwijs (HEO) 2020-2021* (tech. rep.). (2020). Vereniging Hogescholen. Retrieved July 30, 2024, from https://www.vereniginghogescholen.nl/system/knowledge_base/attachments/files/000/001/127/original/Definitieve_taaopdracht_sectorale_verkenning_heo_2020_-_2021.pdf?1594373979
- Talent op de juiste plek* (tech. rep.). (2024, July). IBO doelmatig Hoger onderwijs. Retrieved July 26, 2024, from https://www.tweedekamer.nl/kamerstukken/brieven%5C_regering/detail?did=2024D29131%5C&id=2024Z12068
- Tinto, V. (2012, March). *Completing College: Rethinking Institutional Action*. University of Chicago Press.
- Vervolgonderzoek bevestigt indirecte discriminatie controles uitwonendenbeurs. (2024, May). Retrieved August 1, 2024, from <https://www.duo.nl/particulier/home/actueel/vervolgonderzoek-bevestigt-indirecte-discriminatie-controles-uitwonendenbeurs.jsp>

- Vooren, M., Cornelisz, I., Meeter, M., & Klaveren, C. V. (2024). Bindend studieadvies helpt studenten niet. *Economisch Statistische Berichten*, 4836(109), 351–353.
- Vrijwel geen toezicht op gebruik algoritmes in het onderwijs. (2024, May). Retrieved August 1, 2024, from <https://www.scienceguide.nl/2024/05/vrijwel-geen-toezicht-op-gebruik-algoritmes-in-het-onderwijs/>
- Wat is discriminatie? | Mensenrechten voor jou | College voor de Rechten van de Mens. (n.d.). Retrieved August 18, 2024, from <https://www.mensenrechten.nl/mensenrechten-voor-jou/discriminatie-en-gelijke-behandeling/wat-is-discriminatie>
- Wiśniewski, J., & Biecek, P. (2022). Fairmodels: A flexible tool for bias detection, visualization, and mitigation in binary classification models. *The R Journal*, 14, 227–243. <https://doi.org/10.32614/RJ-2022-019>

Part II

APPENDICES



APPENDIX A - FAIRNESS ANALYSIS PROCESS

A.1 FAIRNESS ANALYSIS

The process of arriving at a prediction model for retention after 1 year and thus calculating fairness consists of three steps: 1) developing prediction models¹, 2) determining the importance of variables, and 3) calculating fairness².

A.1.1 *Step 1 - Developing prediction models*

The first step is to prepare the data and build several prediction models.

- After completing the necessary procedures, we first collect historical study data. For each first-year student in each study program and academic year—the cohort—there is a row in the dataset containing study data. We combine data where necessary and resolve any errors.
- We remove rows without an outcome variable, retention after 1 year³.
- We then resolve rows with missing data, as prediction models cannot handle these. It is important to first investigate missing values thoroughly to understand why they are missing. For example, vocational students and students with foreign prior education do not have an average grade from their prior education. We fill in rows with missing data (imputation). For each variable we fill in, we create a new variable regardless of whether there was an original value or not (VariableName_Missing).

¹ For this study, we used the tidymodels package in R (Kuhn & Wickham, 2020).

² For the research into variables and fairness, we used the DALEX, DALEXtra, and fairmodels packages in R (Biecek, 2018; Maksymiuk, Gosiewska, & Biecek, 2020; Wiśniewski & Biecek, 2022).

³ This occurred in our research for a study programme that had only just started.

- We then examine variables that are strongly correlated with each other. Examples include the average grade for the central final exam and the average grade for the school exam. For strongly correlated variables, we normally retain the most predictive variable, but in this study, we retain all variables so we can understand their relationship afterwards.
- Next, we divide the data into three parts: the training data, the test data, and the validation data. For example, we split the total dataset into 60% for training, 20% for testing, and 20% for validating the model we have developed (Figure A.1). In this division, we ensure that the ratio of students who continue their studies after 1 year is the same everywhere (stratification). How well the model performs on the validation data will determine its quality, such as its accuracy.
- Now we create the prediction model. It is important to create a prediction model for each study programme, because the students who enroll in each study programme are unique. Enrollment depends on the type of study programme (associate degree or bachelor's degree), the form of the study programme (full-time, part-time, or dual), the nature of the study programme (the subject), and, for the geographical distribution of students, the degree of competition in the region. For each study programme, we fit two models for retention after one year: a logistic regression and a random forest. We choose these models because they make it easy to explain which variables are included and what each variable contributes to the prediction.
- Finally, we calculate the quality for each study programme and each model using the Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC). The AUC/ROC measures the extent to which a prediction model correctly classifies positive outcomes as positive and negative outcomes as negative. In our example, this means that the model accurately predicts which students will or will not continue their studies after one year. The model with the highest Area Under the Curve (AUC/ROC) forms the basis for the next step: calculating fairness.

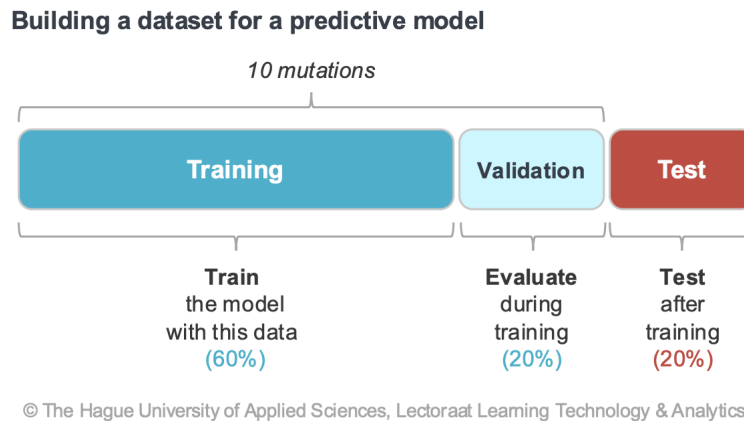


Figure A.1: Division of the dataset into three parts: the training data, the test data, and the validation data

A.1.2 Step 2 - Determining the importance of variables

The second part of the analysis consists of gaining a better understanding of the variables in the prediction model:

- After developing two models for each study programme, we continue with the model that has the highest degree of accuracy. For that model, we identify which variables are important for predicting retention after 1 year and how they contribute to the probability of retention.
- We then apply the models to the average student in a study programme. We calculate the average student by averaging the entire student population within the study programme for continuous variables, and by using the most common value for categorical variables in order of importance within the model. See Figure A.2.
- We then repeat the analysis for different groups. We examine gender, highest prior education, and type of connection⁴. Here, too, we calculate an average student for each group and then apply the prediction model. See Figure A.3.
- To better understand the patterns we find in this, we apply follow-up analyses to assess the ranking of variables, better understand the individual effects of

⁴ Type of connection refers to how the student entered the program after completing their highest prior education: was it directly after prior education, after a gap year, after an internal or external switch, after a colloquium doctum or 21+ test, as a second degree, or is the connection unknown?

variables, and estimate the degree of variance within groups: a Shapley analysis, a Ceteris Paribus analysis (Figure A.4), and a Partial Dependency analysis (Figure A.5).

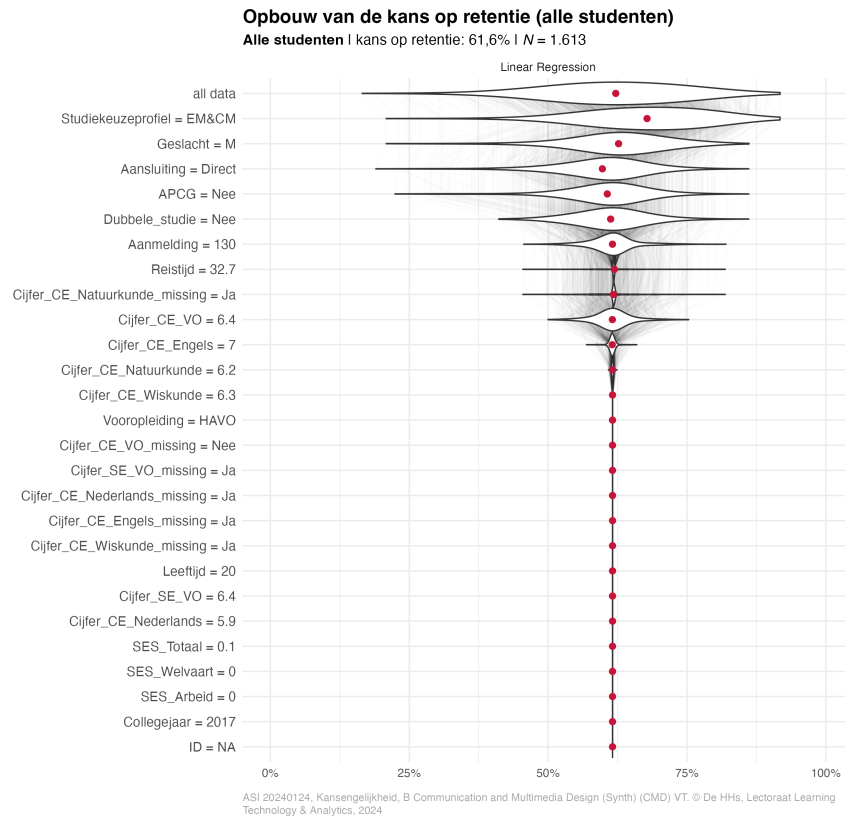
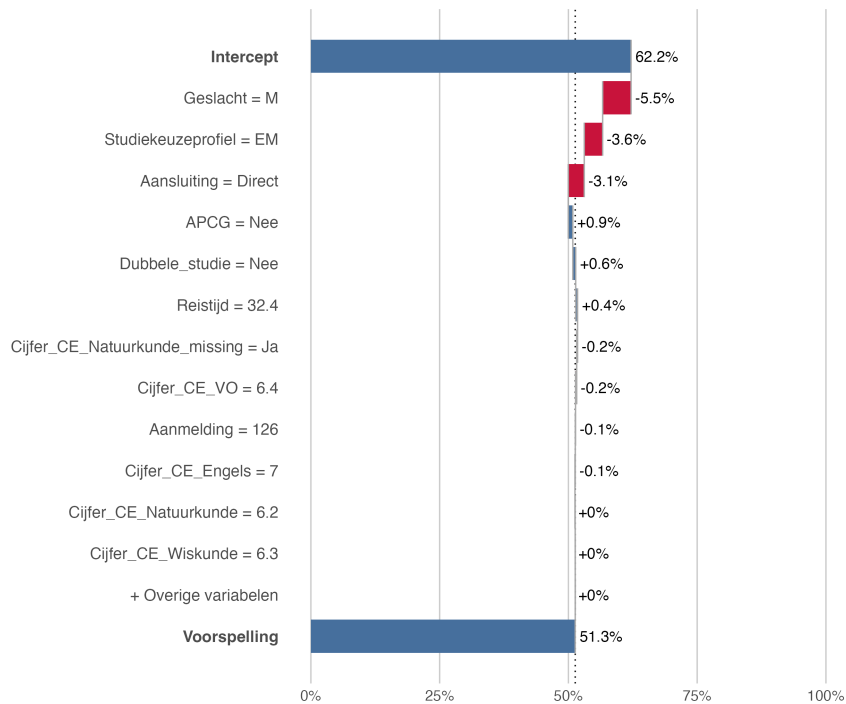


Figure A.2: Application of the model to an average, fictional student

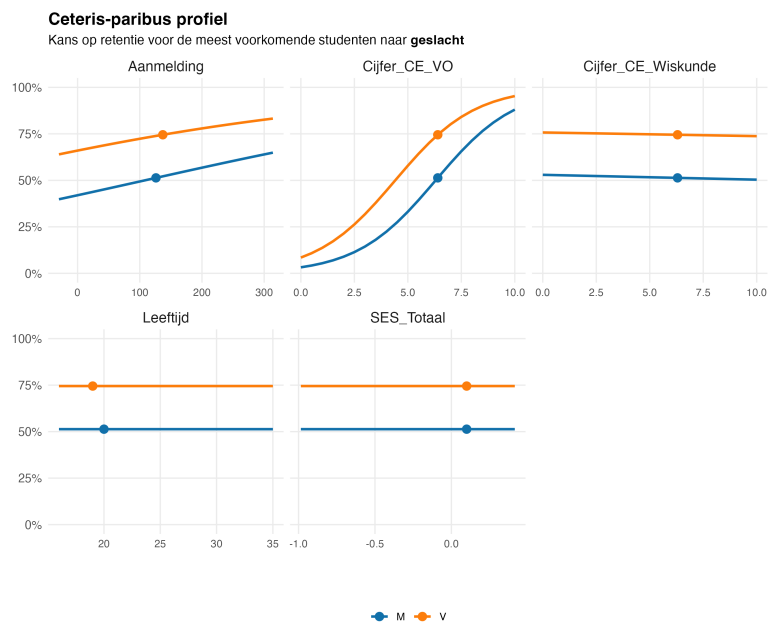
Opbouw van de kans op retentie naar geslacht

Geslacht: M | kans op retentie: 51,3% | N = 915 van 1.613 (56,7%)



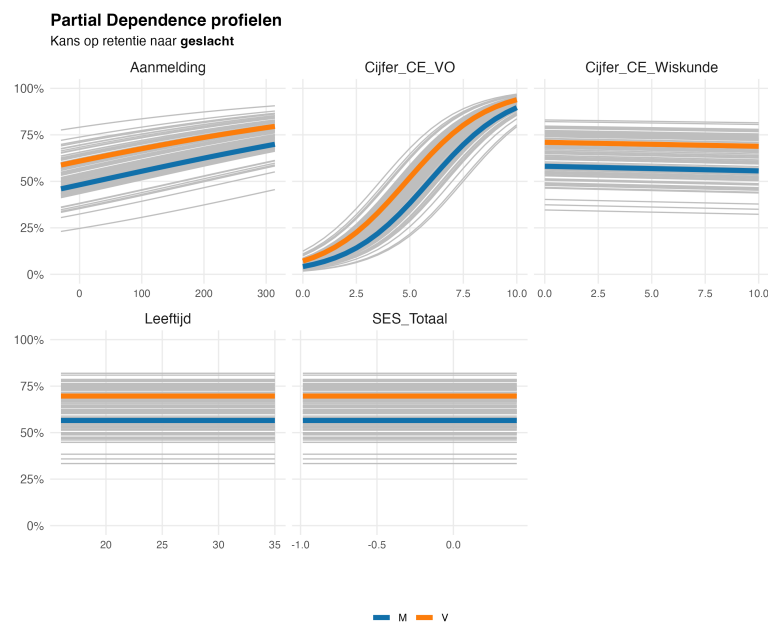
ASI 20240124, Kansengelijkheid, B Communication and Multimedia Design (Synth) (CMD) VT. © De HHs, Lectoraat Learning Technology & Analytics, 2024

Figure A.3: Application of the model to an average, fictional male/female student



ASI 20240124, Kansengelijkheid, B Communication and Multimedia Design (Synth) (CMD) VT. © De HHs, Lectoraat Learning Technology & Analytics, 2024

Figure A.4: Ceteris Paribus Plot



ASI 20240124, Kansengelijkheid, B Communication and Multimedia Design (Synth) (CMD) VT. © De HHs, Lectoraat Learning Technology & Analytics, 2024

Figure A.5: Partial Dependence Plot

A.1.3 Step 3 - Calculating fairness

The third and final part of the process is calculating fairness:

- First, we analyze the probability distribution within groups; this gives us a good picture of the distribution of probabilities in the prediction model for each group and allows us to compare them effectively. See Figure A.6.
- Next, we apply the 4/5 criterion to a number of performance values, including accuracy⁵. As the privileged group, we use the most common subgroup within a variable⁶. If two or more values fall outside the bandwidth of the 4/5 criterion, there is a lack of fairness for the protected group in relation to the privileged group. This can be negative, but also positive for some groups. We also calculate the 'total loss' per variable. The larger this number, the greater the differences within a study programme. See Figure A.7.

⁵ A total of 5 ratios were assessed: Accuracy Equality, Equal Opportunity, Predictive Equality, Predictive Parity, and Statistical Parity. See Appendix A for an explanation of these ratios.

⁶ Suppose there are more men than women in a study programme, then 'Men' is the privileged group; in the event of a tie, one of the two is chosen.

- Finally, we summarize the findings in a table to see where there is bias and for which groups. We distinguish between groups for which we have sufficient data to make a statement and those for which we do not⁷. See Figure A.8.

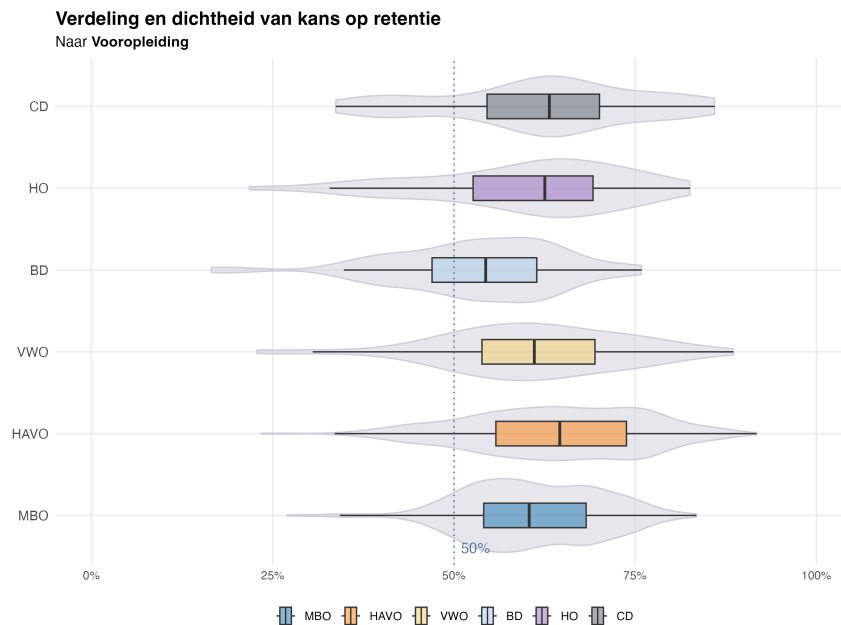


Figure A.6: Distribution of the probability of retention after 1 year by highest pre-training

⁷ NB. It may be that the privileged group is the group that suffers from bias. This can be determined if all other groups have a negative or positive bias

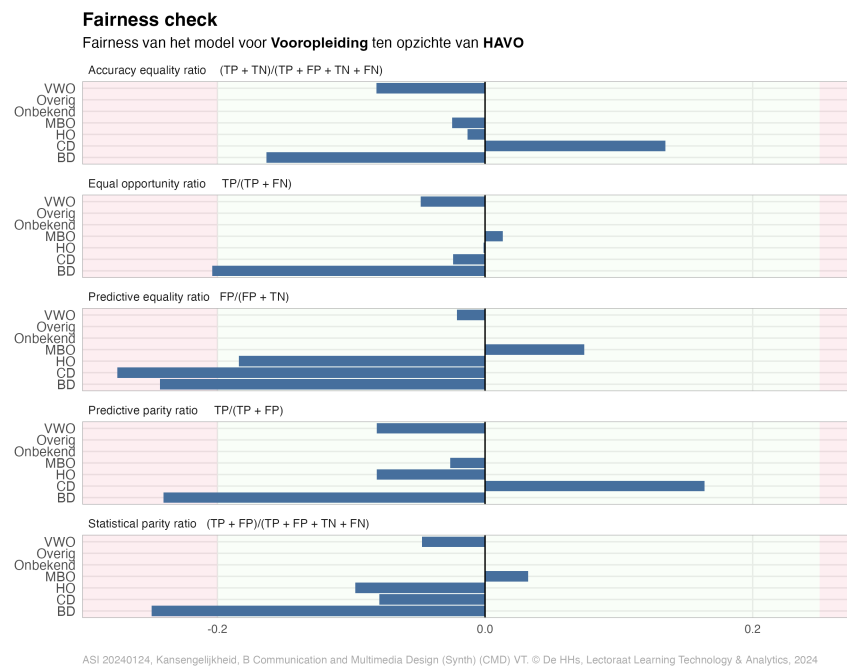


Figure A.7: Fairness analysis by highest level of prior education

Variabele	Groep	Bias	Geen Bias	Negatieve Bias	Positieve Bias
<i>Geslacht</i>	V	Nee	5	0	0
<i>Vooropleiding</i>	BD	Ja	1	3	1
	CD	Ja	0	0	5
	HO	Nee	5	0	0
	MBO	Ja	2	3	0
	VWO	Ja	3	0	2
<i>Aansluiting</i>	2e Studie	Ja	2	2	1
	Na CD	Ja	0	0	5
	Switch extern	Nee	4	0	1
	Switch intern	Nee	4	1	0
	Tussenjaar	Ja	2	0	3

Figure A.8: Fairness conclusions per variable

We have completed this procedure for all associate degree and bachelor's study programmes at The Hague University of Applied Sciences, in all variants. For each study programme, we have incorporated the procedure into a self-explanatory website

with a separate page for each step, including the underlying code. Starting today, the Research Group will make the source code for this method available to anyone in education who wants to apply this approach. Further dissemination will take place via the Npuls program in collaboration with The Hague University of Applied Sciences.

B

APPENDIX B - FAIRNESS RATIOS

For each group, we examine five ratios or metrics. These ratios are derived from relationships in the confusion matrix; they provide insight into the degree of bias and equity from different perspectives of a prediction model.

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	② Sensitivity Equal opportunity ratio (EO) $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		④ Precision Predictive parity ratio (PPV) $\frac{TP}{(TP + FP)}$	Negative Predicted Value $\frac{TN}{(TN + FN)}$	① Accuracy Accuracy equality ratio (ACC) $\frac{TP + TN}{(TP + TN + FP + FN)}$
				⑤ Statistical parity ratio (STP) $\frac{TP + FN}{(TP + TN + FP + FN)}$
				③ Predictive opportunity ratio (PO) $\frac{FN}{(FN + TN)}$

© The Hague University of Applied Sciences, Professorship Learning Technology & Analytics

Figure B.1: Confusion Matrix with 5 ratios for fairness

B.1 ACCURACY EQUALITY

To what extent does the model accurately predict both positive and negative outcomes?

This metric assesses whether a model is equally accurate across different subgroups within the dataset. It compares the model's accuracy for a protected group (a minority group) with that for a privileged group (the majority group). This ratio is calculated as the ratio of the accuracy for the protected group to that for the privileged group.

Formula

$$ACC = \frac{TP + TN}{TP + FP + TN + FN}$$

Definitions

- *Accuracy*: The percentage of correct predictions made by the model. This is calculated as the number of correct predictions divided by the total number of predictions.

Interpretation

- *Ratio = 1*: The model is equally accurate for both groups.
- *Ratio < 1*: The model is less accurate for the protected group than for the privileged group, indicating possible bias against the protected group.
- *Ratio > 1*: The model is more accurate for the protected group than for the privileged group, which may also indicate a form of bias, but in favor of the protected group.

Example

Suppose we predict retention for HAVO students (privileged) and MBO students (protected), and the model's accuracy is 80% for HAVO students and 70% for MBO students. The Accuracy Equality Ratio is then 0.875 (70% / 80%). This means that the model is less accurate for the protected group (vocational students) than for the privileged group (HAVO students).

B.2 EQUAL OPPORTUNITY

To what extent are the correctly positive predictions equal?

This metric assesses the equality of opportunity that a model offers to different subgroups in terms of the 'True Positive Rate' (TPR). It compares the probability that a

model correctly predicts a positive outcome for a protected group versus a privileged group.

The True Positive Rate (TPR) is also referred to as *sensitivity* (*sensitivity* or *recall*). The Equal Opportunity Ratio is calculated as the number of true positives divided by the total number of actual positives.

Formula

$$\text{TPR} = \frac{TP}{TP + FN}$$

Definitions

- *TP*: True Positives (correct predictions of positive outcomes)
- *FN*: False Negatives (actual positive outcomes that are incorrectly predicted as negative)

Interpretation

- *Ratio = 1*: The model offers equal opportunities to both groups in terms of correctly predicting positive outcomes.
- *Ratio < 1*: The model offers fewer opportunities to the protected group compared to the privileged group, indicating possible bias against the protected group.
- *Ratio > 1*: The model offers more opportunities to the protected group compared to the privileged group, which may also indicate a form of bias, but in favor of the protected group.

Example

Suppose we predict retention for HAVO students (privileged) and MBO students (protected), and the True Positive Rates are 70% for HAVO students and 60% for MBO students. The Equal Opportunity Ratio is then 0.857 (60% / 70%). This means the model offers the protected group (MBO students) a lower chance of correctly predicting positive outcomes than the privileged group (HAVO students).

B.3 PREDICTIVE EQUALITY

To what extent are the false positive predictions equal?

This metric assesses the equality of the number of False Positives between different subgroups. It compares the False Positive Rate (FPR) for a protected group with the FPR for a privileged group.

The FPR is the ratio of the number of false positive predictions (False Positives, FP) to the total number of actual negative cases (True Negatives, TN, and False Positives, FP). The Predictive Equality Ratio is calculated as the ratio of the FPR for the protected group to that for the privileged group.

Formula

$$\text{FPR} = \frac{FP}{FP + TN}$$

Definitions

- *FP*: False Positives (incorrect predictions of positive outcomes)
- *TN*: True Negatives (correct predictions of negative outcomes)

Interpretation

- *Ratio = 1*: The model has an equal chance of producing False Positives for both groups.
- *Ratio < 1*: The model is less likely to produce False Positives for the protected group than for the privileged group, indicating possible bias against the privileged group.
- *Ratio > 1*: The model is more likely to produce false positives for the protected group than for the privileged group, indicating possible bias against the protected group.

Example

Suppose we predict retention for HAVO students (privileged) and MBO students (protected), and the False Positive Rate for HAVO students is 20% and for MBO students 30%. The Predictive Equality Ratio is then 1.5 (30% / 20%). This means the model is more likely to make false-positive predictions for the protected group (MBO students) than for the privileged group (HAVO students).

B.4 PREDICTIVE PARITY

To what extent are the justified positive predictions equal?

This metric assesses the equality of predictive accuracy across subgroups by comparing the positive predictive value (PPV) of the model for a protected group with that for a privileged group. The PPV, also known as *precision*, is calculated as the number of true positives (TP) divided by the total number of predicted positives (TP + FP). It is a measure of the model's accuracy in positive predictions.

Formula

$$\text{PPV} = \frac{TP}{TP + FP}$$

Definitions

- *TP*: True Positives (correct predictions of positive outcomes)
- *FP*: False Positives (incorrect predictions of positive outcomes)

Interpretation

- *Ratio* = 1: The model has equal accuracy in predictions for both groups.
- *Ratio* < 1: The model is less accurate in predicting positive outcomes for the protected group than for the privileged group, indicating possible bias against the protected group.
- *Ratio* > 1: The model is more accurate in predicting positive outcomes for the protected group than for the privileged group, which may also indicate a form of bias, but in favor of the protected group.

Example

Suppose we predict retention for high school graduates (privileged) and vocational school graduates (protected), and the Positive Predictive Value for high school graduates is 80% and for vocational school graduates 70%. The Predictive Parity Ratio is then 0.875 (70% / 80%). This means that the model is less accurate in predicting positive outcomes for the protected group (vocational students) than for the privileged group (HAVO students).

B.5 STATISTICAL PARITY

To what extent are the positive predictions equal?

This measure assesses the equality of positive predictions (whether correct or not) across different subgroups. It compares the probability that a model predicts a positive outcome for a protected group with the probability that it predicts a positive outcome for a privileged group. The Statistical Parity Ratio is the ratio of the probability of a positive prediction for the protected group to that for the privileged group.

Formula

$$\text{SPR} = \frac{TP + FP}{TP + FP + TN + FN}$$

Definitions

- *Positive Prediction*: A prediction in which the model predicts a positive outcome (e.g., acceptance, credit approval, etc.).

Interpretation

- *Ratio = 1*: The model predicts positive outcomes equally often for both groups.
- *Ratio < 1*: The model predicts positive outcomes less often for the protected group than for the privileged group, indicating possible bias against the protected group.
- *Ratio > 1*: The model predicts positive outcomes more often for the protected group than for the privileged group, which may indicate bias against the privileged group.

Example

Suppose we predict retention for HAVO students (privileged) and MBO students (protected), and the probability of a positive prediction for HAVO students is 80% and for MBO students 70%. The Statistical Parity Ratio is then 0.875 (70% / 80%). This means that the model predicts positive outcomes less often for the protected group (HAVO students) than for the privileged group (MBO students).

TYPOGRAPHY

This document has been formatted with the typographic look and feel `classicthesis` developed by

André Miede and Ivo Pletikosić: <https://bitbucket.org/amiede/classicthesis/>. The cover was developed using a Latex template from Delft University of Technology.

NO FAIRNESS WITHOUT AWARENESS

Applied Research into Equity in Higher Education Inaugural Address upon Appointment as Professor of Learning Technology & Analytics

In his inaugural address, Theo C. Bakker, PhD, discusses equity in higher education and the role of data analysis in promoting social justice. From his Research Group for Learning Technology & Analytics at The Hague University of Applied Sciences, Bakker conducts research on equity across different student groups, using predictive models and study data to identify structural inequalities. He emphasizes that equity can only be resolved by first consciously examining the education system. *No Fairness without Awareness*.

Learning Analytics and bias – Learning analytics focuses on measuring and analyzing student data to improve education. Bakker distinguishes between levels, such as student and institutional analytics, and focuses on inclusion analytics, which examines equity. Bias-systematic prejudices in data can reinforce prejudices in algorithms and thus cause inequality of opportunity. The research method uses the 4/5 criterion, in which fairness in outcomes is measured by examining whether opportunities for the protected group are at least 80% of those for the privileged group.

Research approach – Bakker uses machine learning to predict dropout rates after the first year of study, then examines differences between groups of students, such as vocational and pre-university students. He follows three steps: (1) *Prepare data and build models*: Data is split and cleaned to create accurate prediction models. (2) *Analyzing variables*: The influence of characteristics on the results is assessed for different groups. (3) *Calculating fairness*: The 4/5 criterion is applied to metrics such as accuracy and statistical equality to identify bias and inequalities.

Results, recommendations, and follow-up research – The study shows that many study programmes lack equal opportunities, particularly for men and vocational students, who are more likely to drop out. Bakker recommends including sensitive characteristics such as migration background in analyses based on informed consent. He also advocates greater policy flexibility, inspired by measures taken during the coronavirus crisis that had a positive effect on study success.

Vision for the future – Bakker emphasizes that not all inequality stems from discrimination and calls for data-informed interventions to promote social justice in education. His method is made available through open access so that other institutions can also apply it and systematically and consciously investigate equity.